NORTH-WEST UNIVERSITY · CMPG 325 COMPUTER NETWORKS

Technical Report

Design, Implementation and Verification of a Secured Network for the Katlego Adult Learning Centre

Individual Semester Project · Final Submission

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Project ID / Client ID	CMPG325-2026-025 / CLI-025
Client	Katlego Adult Learning Centre, Rustenburg, North West
Assigned technical challenge	Port Security (switchport access control)
Assigned address block	192.168.22.0/24
Change request	CR11: adoption of VoIP handsets
Simulation tool	Cisco Packet Tracer
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Repository: `https://github.com/konyanalethabo-work/katlego-network-project`

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1 Executive summary

This report documents the design, implementation and verification of a computer network for the Katlego Adult Learning Centre, a small community adult-education facility in Rustenburg. It consolidates the Client Design Review (Milestone 1) and the Client Implementation Review (Milestone 2) into one technical record and adds the work completed for the final submission: hardening of every unused switch port, and a simulated Internet edge with a DNS server.

The network is segmented into five VLANs, routed by a single router using router-on-a-stick, built on a core switch and three access switches, and addressed from the assigned block `192.168.22.0/24` with Variable Length Subnet Masking (VLSM). The assigned technical challenge, **Port Security**, is enforced on all 21 access ports and was demonstrated by causing a real violation and recovering from it.

Results at a glance

Segments	5 VLANs (10, 20, 30, 40, 99), one subnet each
Addressing	VLSM; 80 of 256 addresses allocated (31.25%), 176 held in reserve
End devices served	19 PCs, 1 shared printer, 1 VoIP handset
Port Security	21 of 21 access ports secured (sticky MAC, violation action `shutdown`)
Unused ports disabled	76 ports across the four switches
Server-room protection	ACL 100 blocks VLANs 10, 20 and 40; Admin (VLAN 30) permitted
External connectivity	Default route to a simulated ISP; DNS server at 8.8.8.8
Tests performed	17, all passed (Section 7)

1.1 How this report maps to the marking rubric

Rubric criterion	Weight	Where it is addressed
1. Client requirements and network design	15%	Sections 2, 3 (topologies, VLSM plan, justified decisions)
2. Packet Tracer implementation and functionality	35%	Sections 4, 6, 7; Appendix A
3. Assigned feature: Port Security	20%	Section 5; tests T5, T8, T17
4. GitHub portfolio of evidence	15%	Section 10
5. Video demonstration and technical defence	15%	Section 10 (companion deliverable)

2 Client background and requirements analysis

2.1 The client

Katlego Adult Learning Centre is a community-based adult education facility in Rustenburg, North West Province, the centre of South Africa's platinum mining industry. Many learners are mineworkers and their families who had limited access to formal schooling. The centre offers literacy, numeracy and basic digital skills. It is a modest, single-site institution with a limited budget, a small staff and learner base, and **no dedicated IT staff on site**. These facts drive the design philosophy: the network must be affordable, simple to operate, and secure by default.

2.2 Project scope

As defined in the client brief (CMPG325-2026-025), the network must:

- provide connectivity and network services across the facility;
- use the assigned addressing block `192.168.22.0/24`;
- implement the assigned technical challenge: Port Security (switchport access control);
- respect the design constraint that access to the server room is restricted;
- absorb Change Request CR11 (VoIP handsets) without adding scope beyond what is required.

2.3 Facility and device inventory

The facility is modelled as a small single-storey building of seven functional spaces, the smallest layout that still demonstrates distinct traffic types: classroom, laboratory, administration, voice and restricted infrastructure.

Room	Function	Devices
Classroom A	General instruction (literacy, numeracy)	1 teacher PC
Classroom B	Second subject stream	1 teacher PC
Computer Lab	Digital-literacy training; highest device density	11 PCs
Admin Office	Registration, finance, management	3 PCs, 1 printer
Reception	First point of contact	1 PC, 1 VoIP handset
Library / Resource Room	Shared self-study computers	2 PCs
Server Room	Core network infrastructure; access-restricted	Router, core switch

In total the facility must serve **19 PCs, 1 printer and 1 VoIP handset**, with headroom for further handsets as anticipated by CR11. Device counts were kept lean to avoid over-provisioning a client with a limited operating budget.

2.4 Functional requirements

Milestone 1 derived six functional requirements from the client background. A seventh was added at the final stage, because the marking rubric lists DNS and Internet access among the core services a working network must provide.

ID	Requirement	Statement
FR1	Connectivity	All end-user devices must reach appropriate network resources and, where required, each other.
FR2	Segmentation	Traffic is logically separated by function, reflecting different trust levels and traffic types.
FR3	Scalable addressing	Efficient use of `192.168.22.0/24` through VLSM, sized per segment with room for growth.
FR4	Port-level access control	Port Security prevents unauthorised devices from connecting, given the limited on-site IT oversight.
FR5	Restricted infrastructure	The server room and infrastructure are isolated; access is limited to authorised staff.
FR6	Voice readiness (CR11)	VoIP handsets are carried on a dedicated voice segment, avoiding contention with data.
FR7	External access and DNS	Devices reach an external network and resolve names through a DNS server (added at the final stage).

2.5 Assumptions

1. **Authorised staff.** The centre has no IT staff on site, so the Admin Office (VLAN 30) stands in for "authorised staff" and is deliberately left unrestricted towards the server room. Milestone 1 did not state this; Milestone 2 introduced it, and it is retained here.
2. **Single site.** One building, one router, one Internet exit.
3. **Simulated Internet.** The Internet is represented by one ISP router and one DNS server inside Packet Tracer. No address translation is performed; the ISP carries a static return route to the client block (Section 4.6).
4. **Phone placement.** One VoIP handset is deployed at Reception. A single handset is enough to justify a dedicated voice VLAN while leaving deliberate headroom for later rollout.

3 Network design

3.1 Design decisions and their justification

Every significant decision is stated once, applied consistently, and traced to a requirement.

ID	Decision	Justification	Traces to
D1	DHCP on user VLANs; static addresses on VLAN 99; fixed printer address	With no IT staff on site, nobody should type addresses into twenty devices. The printer is addressed statically (192.168.22.50) because others print to it by address. VLAN 99 is static so management never depends on a service reached through the network it manages.	FR1, FR3
D2	Five VLANs grouped by function	Simple enough to implement and defend, while reflecting industry practice: learning spaces, lab, admin, voice, infrastructure.	FR2
D3	Router-on-a-stick	One physical trunk and five sub-interfaces suffice for 21 devices and keep hardware cost low.	FR1, FR2
D4	Pruned trunks	Each uplink carries only the VLANs its switch needs, so broadcasts never reach a switch with no member devices.	FR2
D5	Port Security: sticky MAC, maximum 1 (2 on the phone port), violation action `shutdown`	A port that quietly drops traffic goes unnoticed without IT staff; a port that visibly goes down gets fixed. It also gives unambiguous evidence.	FR4
D6	Disable every unused port	Port Security protects ports in use. An empty open port is an unprotected door; shutting it removes the opportunity entirely.	FR4
D7	ACL on the router blocks VLANs 10, 20, 40 from the VLAN 99 subnet	A separate subnet alone does not isolate: with routing enabled every VLAN could still reach it.	FR5
D8	Simulated ISP, default route and return route, DNS server at 8.8.8.8	Provides the Internet and DNS services expected of a working network. 8.8.8.8 is the DNS address DHCP already hands out, so no PC needs reconfiguring.	FR7

3.2 Segmentation

VLAN	Segment	Justification
10	Classrooms and Library	Low-density learning-support spaces with similar, light traffic needs
20	Computer Lab	Highest device density; its own broadcast domain
30	Admin and Reception	Business operations, different from learner traffic
40	Voice	Dedicated segment for VoIP handsets (CR11), no contention with data
99	Management / Server room	Isolated infrastructure segment satisfying the restricted-access constraint

3.3 Physical topology

Figure 1 shows the devices and cabling. The router and core switch sit in the access-restricted server room. Three access switches serve the remaining rooms, grouped to match the VLAN structure. The simulated Internet (ISP router and DNS server) is attached to the router's second Gigabit port.

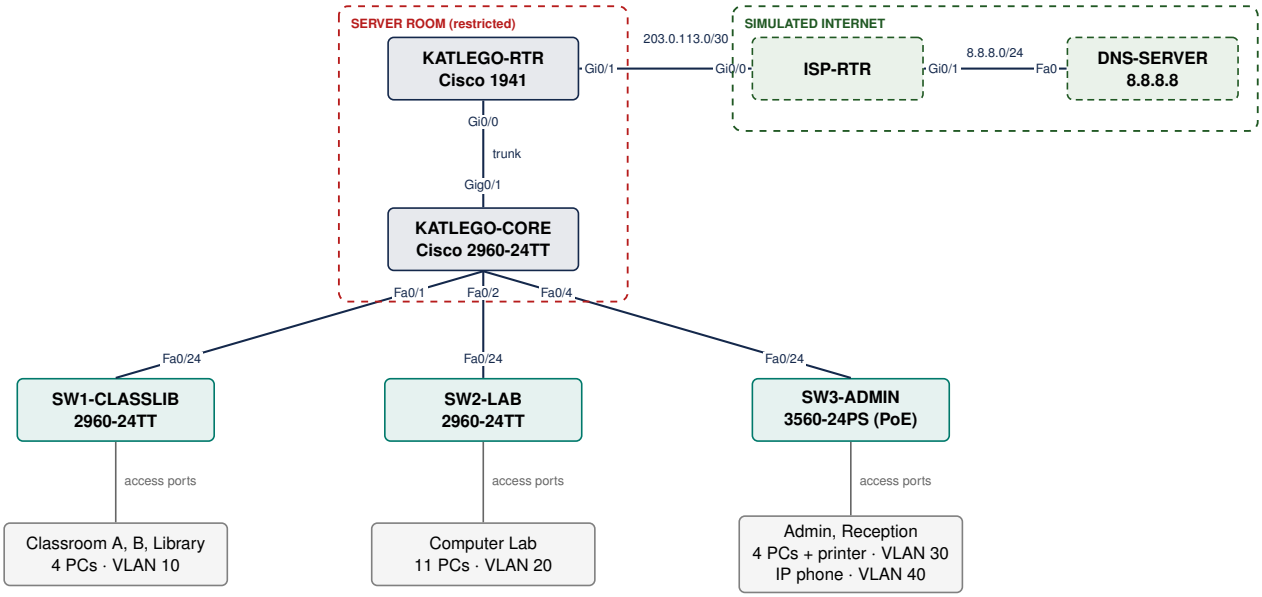


Figure 1: *Physical topology. Solid navy lines are trunk or routed links; dashed green devices are the simulated Internet added at the final stage.*

Cabling summary. The router connects to the core switch by a single trunk. Core-to-access links are trunks carrying tagged VLAN traffic, pruned to the VLANs each switch needs. Access-switch-to-device links are access-mode ports, each secured individually with Port Security.

3.4 Logical topology

Figure 2 shows how traffic flows between VLANs, independent of physical location. The router uses one sub-interface per VLAN over the single trunk. ACL 100 is applied inbound on the sub-interfaces of VLANs 10, 20 and 40; VLAN 30 is the deliberate exception.

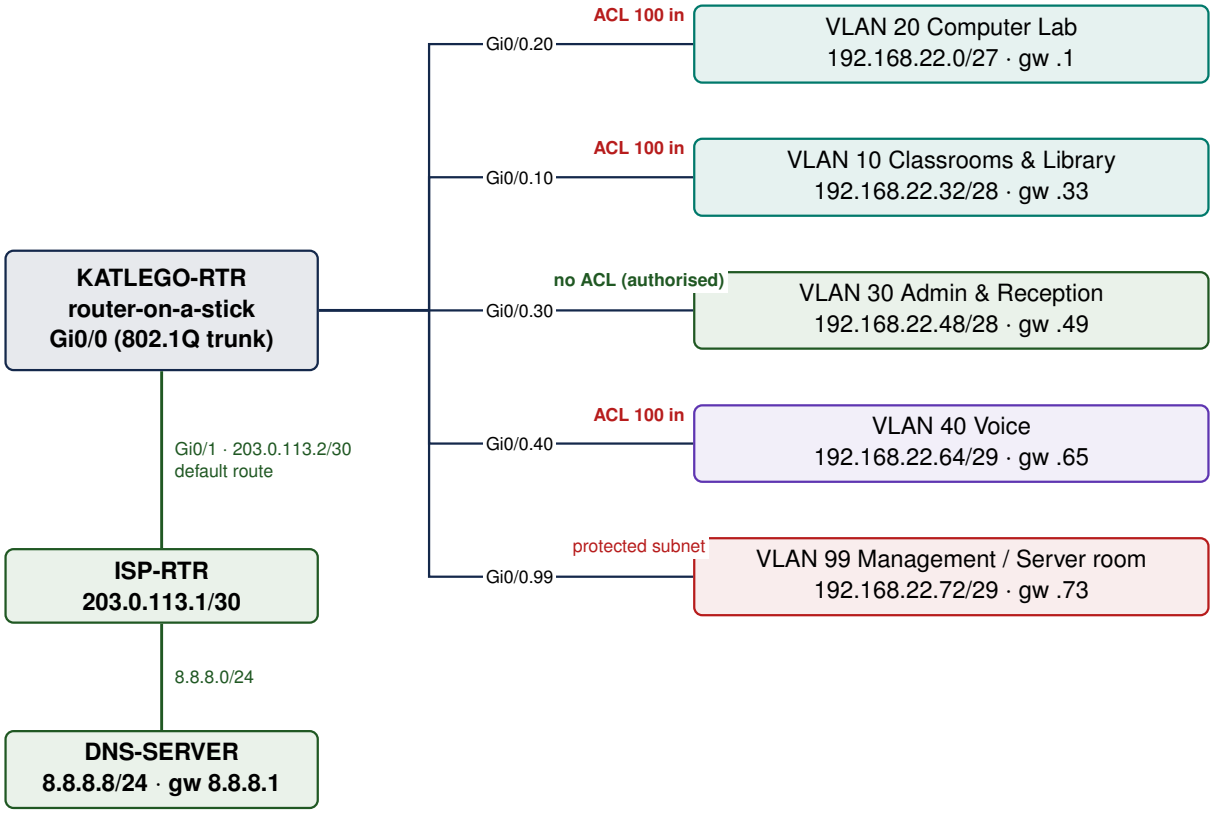


Figure 2: *Logical topology: VLAN-to-subnet mapping through router sub-interfaces, ACL placement and the Internet edge.*

3.5 IP addressing plan

3.5.1 Method

The client holds a single block, `192.168.22.0/24` (256 addresses, 254 usable). Instead of equal fixed-size subnets, VLSM sizes each subnet to the segment it serves plus headroom. Subnets are allocated **largest first**, which avoids fragmentation of the remaining space and simplifies later recalculation. Each subnet's first usable address is its default gateway.

3.5.2 Worked calculation

For a subnet that must hold H hosts, the number of host bits n is the smallest integer with $2^n - 2 \geq H$; the prefix is $32 - n$.

Worked example: VLAN 20 (Computer Lab)

1. **Requirement.** 11 devices now; sized for 30 hosts to allow growth.
2. **Host bits.** $2^4 - 2 = 14 < 30$ and $2^5 - 2 = 30 \geq 30$, so $n = 5$.
3. **Prefix.** $32 - 5 = 27$, mask 255.255.255.224.
4. **Block size.** $2^5 = 32$ addresses. First block starts at `192.168.22.0`.
5. **Range.** Network `.0`, usable `.1`–`.30`, broadcast `.31`; gateway `.1`.

Answer: **192.168.22.0/27** usable **.1 – .30** broadcast **.31**

The same five steps applied to every segment, in allocation order:

VLAN	Hosts now	Sized for	n	Prefix	Network	Usable range
20 Computer Lab	11	30	5	**/27**	192.168.22.0	.1 – .30
10 Classrooms & Library	4	14	4	**/28**	192.168.22.32	.33 – .46
30 Admin & Reception	5	14	4	**/28**	192.168.22.48	.49 – .62
40 Voice	1	6	3	**/29**	192.168.22.64	.65 – .70
99 Management	5	6	3	**/29**	192.168.22.72	.73 – .78
Router–ISP link	2	2	2	**/30**	203.0.113.0	.1 – .2

The router–ISP link needs exactly two addresses, so a `/30` is the correct size. The `203.0.113.0/24` range is reserved for documentation and examples, which makes it safe to use in a simulation.

Allocation check

1. Block sizes: $32+16+16+8+8=80$ addresses, allocated contiguously from `.0` to `.79`.
2. Allocated share: $80/256 = 31.25\%$.
3. Unallocated: `192.168.22.80` – `.255`, $256-80=176$ addresses (68.75%).

Answer: **80 allocated** **176 reserved for growth**

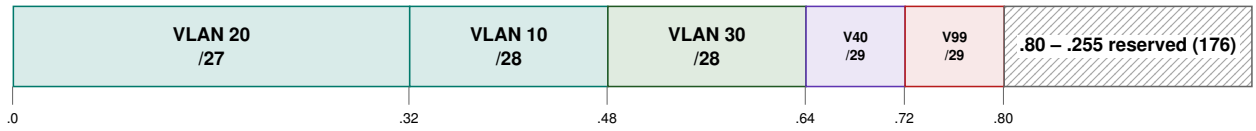


Figure 3: *Address map of `192.168.22.0/24`: the first 80 addresses drawn to scale, the remaining 176 shown as one reserved block.*

3.5.3 Address assignment

Item	Address	Notes
Router sub-interfaces	.33 (VLAN 10), .1 (20), .49 (30), .65 (40), .73 (99)	Gateways, first usable address of each subnet
Management (VLAN 99)	Core .74, SW1 .75, SW2 .76, SW3 .77	Static; one spare address (.78)
Shared printer	192.168.22.50	Static, mask 255.255.255.240, gateway .49
User VLANs	DHCP pools, DNS 8.8.8.8	Gateways and printer excluded from the pools
Voice VLAN	DHCP with option 150	Phone leased 192.168.22.66
Router–ISP link	.2 (KATLEGO-RTR Gi0/1), .1 (ISP-RTR Gi0/0)	203.0.113.0/30
DNS server	8.8.8.8/24, gateway 8.8.8.1	ISP-RTR Gi0/1 is 8.8.8.1

4 Implementation in Cisco Packet Tracer

The network was built bottom-up, because each layer depends on the one below: the switches learn the VLANs, the router connects them, DHCP addresses the devices, the phone is added, and finally the Internet edge is attached. The complete device configurations are in Appendix A; this section shows the key parts and the evidence for each.

4.1 Equipment

Role	Model	Why
Router (KATLEGO-RTR)	Cisco 1941	Sub-interfaces, ACLs and DHCP on one device; two Gigabit ports (LAN trunk, ISP link)
Core, SW1, SW2	Cisco 2960-24TT	Layer 2 switching only; no PoE needed
SW3 (Admin)	Catalyst 3560-24PS	PoE, so it can power the IP phone
IP phone	Cisco 7960	Standard Packet Tracer VoIP endpoint
End devices	19 PCs, 1 printer	Milestone 1 inventory
ISP-RTR	Cisco 1941	Simulated provider edge
DNS-SERVER	Server-PT	Provides DNS for `www.katlego.edu`

Deviation from plan. A Catalyst 3560-24PS replaced the planned 2960-24PC at SW3. It is PoE-capable and powers the phone, so the requirement is met. The phone (Fa0/6) and the reception PC (Fa0/5) sit on separate ports rather than sharing one, and SW3 uplinks to core Fa0/4 (core Fa0/3 is spare).

4.2 Layer 2: VLANs and trunks

Each access switch knows only the VLANs it carries, plus VLAN 99 for management. Its uplink trunk is pruned to the same list, and only the router link carries all five VLANs (decision D4).

KATLEGO-SW1-CLASSLIB (excerpt)

```
vlan 10
 name DATA_CLASSROOMS_LIBRARY
vlan 99
 name MANAGEMENT
interface vlan 99
 ip address 192.168.22.75 255.255.255.248
 no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 4
 switchport mode access
 switchport access vlan 10
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,99
```

```
KATLEGO-SW1-CLASSLIB>enable
KATLEGO-SW1-CLASSLIB#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                                Gig0/2
10   DATA_CLASSROOMS_LIBRARY          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
99   MANAGEMENT                       active
```

Figure 4: *SW1: VLAN 10 holds Fa0/1–Fa0/4 and VLAN 99 exists (Milestone 2 evidence).*

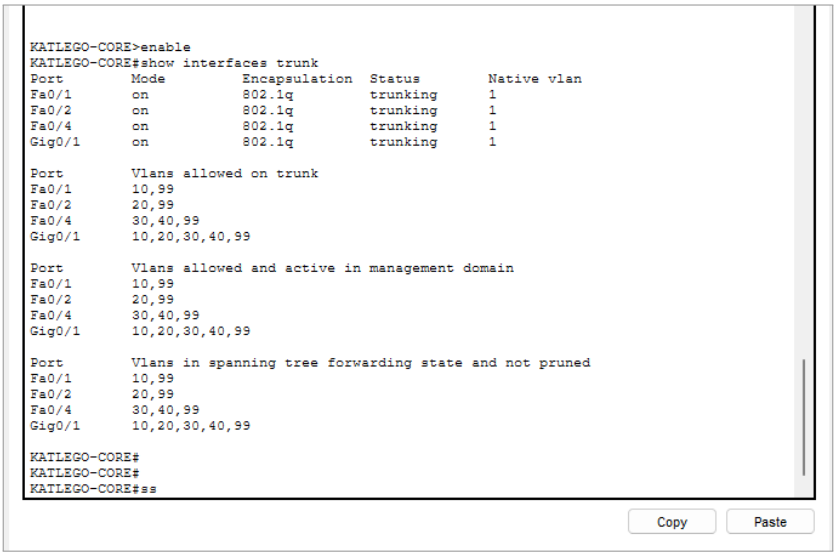


Figure 5: *Core switch: each trunk carries only its pruned VLAN list; Gig0/1 to the router carries all five (Milestone 2 evidence).*

4.3 Layer 3: routing between VLANs

Switches keep VLANs apart; the router lets them talk. One physical link carries tagged traffic and one sub-interface per VLAN acts as that VLAN's gateway.

KATLEGO-RTR: router-on-a-stick

```
interface gigabitEthernet0/0
 no shutdown
interface gigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.22.33 255.255.255.240
interface gigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.22.1 255.255.255.224
interface gigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.22.49 255.255.255.240
interface gigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.22.65 255.255.255.248
interface gigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 192.168.22.73 255.255.255.248
```

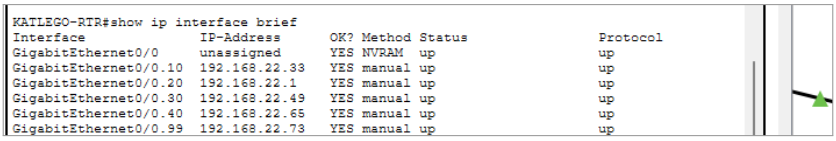


Figure 6: *Router: Gi0/0 and its five sub-interfaces up and addressed (Milestone 2 evidence; the final state is in Figure 10).*

4.4 Addresses: DHCP

Gateways and the printer are excluded from the pools, so they can never be handed out. There is no pool for VLAN 99, which is static by decision D1.

KATLEGO-RTR: DHCP (excerpt)

```
ip dhcp excluded-address 192.168.22.1
ip dhcp excluded-address 192.168.22.33
ip dhcp excluded-address 192.168.22.49 192.168.22.50
ip dhcp excluded-address 192.168.22.65
ip dhcp pool LAB_VLAN20
 network 192.168.22.0 255.255.255.224
 default-router 192.168.22.1
 dns-server 8.8.8.8
ip dhcp pool VOICE_VLAN40
 network 192.168.22.64 255.255.255.248
 default-router 192.168.22.65
 option 150 ip 192.168.22.65
```

The pools for VLAN 10 and VLAN 30 follow the same pattern (Appendix A). Twenty leases were issued: 11 in the Lab, 4 in Classrooms and Library, 4 in Admin and Reception, and 1 voice.

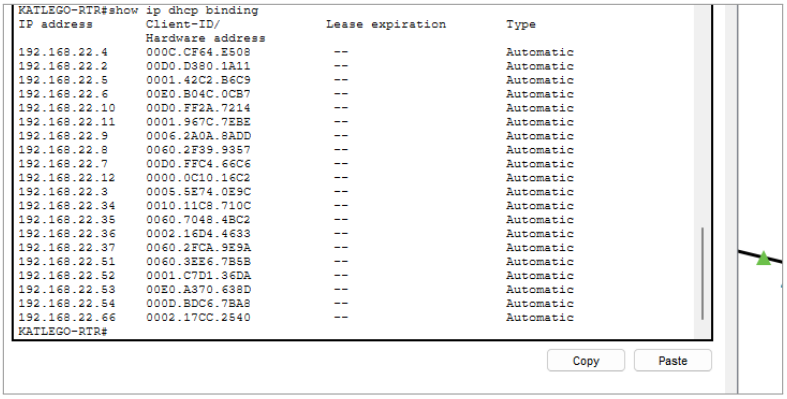


Figure 7: *DHCP bindings on the router: 20 leases, all inside their VLAN ranges, none excluded (Milestone 2 evidence, test T6).*

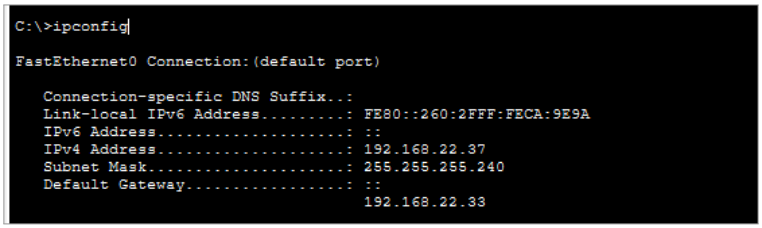


Figure 8: *PC0 holds an address from the VLAN 10 pool, with the correct mask and gateway (Milestone 2 evidence, test T6).*

4.5 Voice on the reception desk

The phone sits on Fa0/6 of the PoE switch, which powers it. The port carries the data VLAN (30) and a voice VLAN (40), so the phone's traffic is tagged into the voice segment. Option 150 in the voice pool tells the phone where to find its call server. This satisfies FR6 and CR11.

KATLEGO-SW3-ADMIN: phone port

```
interface fastEthernet0/6
 switchport mode access
 switchport access vlan 30
 switchport voice vlan 40
```

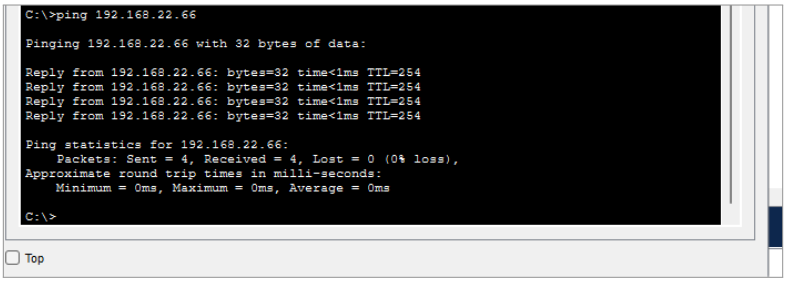


Figure 9: *An Admin PC reaches the IP phone (192.168.22.66) in the voice VLAN: 4 of 4 replies, TTL 254 (Milestone 2 evidence, test T7).*

4.6 Internet edge: ISP router and DNS server

The rubric lists DNS and Internet access among the core services. Although DHCP already advertised the DNS server 8.8.8.8, nothing answered at that address and the network had no external connection. The final stage therefore adds a small simulated Internet (decision D8): an ISP router and a DNS server, attached to the router's Gi0/1 port with Copper Cross-Over cables (router-to-router and router-to-server links join devices of the same kind).

Device	Interface	Address	Connected to
KATLEGO-RTR	Gi0/1	203.0.113.2/30	ISP-RTR Gi0/0
ISP-RTR	Gi0/0	203.0.113.1/30	KATLEGO-RTR Gi0/1
ISP-RTR	Gi0/1	8.8.8.1/24	DNS-SERVER Fa0
DNS-SERVER	Fa0	8.8.8.8/24, gateway 8.8.8.1	ISP-RTR Gi0/1

KATLEGO-RTR: edge link and default route

```
interface gigabitEthernet0/1
 description Link to ISP
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

ISP-RTR

```
hostname ISP-RTR
interface gigabitEthernet0/0
 description Link to Katlego
 ip address 203.0.113.1 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 description Link to DNS server
 ip address 8.8.8.1 255.255.255.0
 no shutdown
exit
ip route 192.168.22.0 255.255.255.0 203.0.113.2
```

On DNS-SERVER the interface is set statically to `8.8.8.8/255.255.255.0` with default gateway `8.8.8.1`, and the DNS service is switched on with one A record: `www.katlego.edu` → `8.8.8.8`.

Why two routes

The **default route** on KATLEGO-RTR reads “for any address I do not know, send it to the ISP”. The centre has one exit, so one default route is the simplest correct choice. The **return route** on ISP-RTR reads “to reach the Katlego block, send it to KATLEGO-RTR”. Without it a request from a PC would reach the server but the reply would have nowhere to go. A real provider would instead rely on address translation, which is outside the scope of this simulation (Section 9).

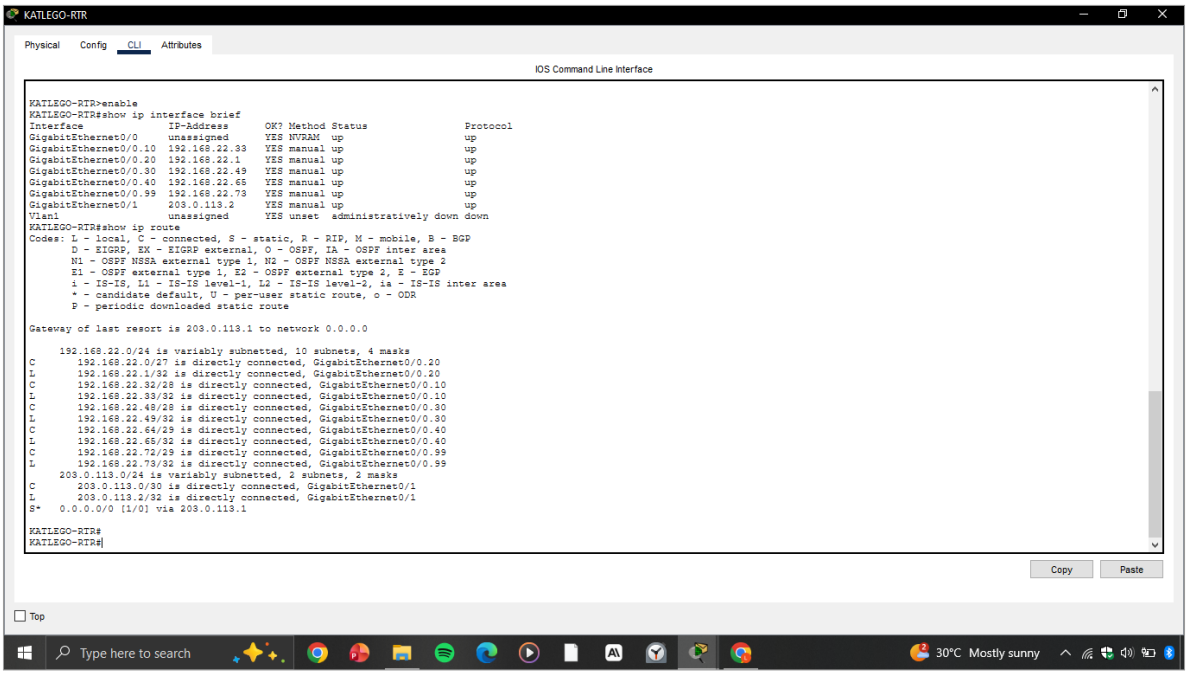


Figure 10: *KATLEGO-RTR final state.* `show ip interface brief`*: Gi0/1 holds 203.0.113.2 and is up/up beside the five sub-interfaces.* `show ip route`*: the gateway of last resort is 203.0.113.1 and the static default route* `S* 0.0.0.0/0 [1/0] via 203.0.113.1` *is installed (tests T9).*

5 Assigned technical challenge: Port Security

5.1 What it is and why it matters here

Every network card has a hardware (MAC) address. Port Security lets a switch limit which MAC addresses may use each port. Without it, anyone can plug a laptop into any wall socket and join the network. The client has no IT staff on site to notice, so the switch itself must refuse unknown devices (FR4).

Violation mode	Behaviour	Suitability for this client
`protect`	Drops traffic from unknown MACs; no log, no counter	Silent. The problem would go unnoticed.
`restrict`	Drops traffic, logs the event, counts violations	Visible only to someone reading logs.
`shutdown`	Port goes err-disabled, event is logged and counted	**Chosen.** The failure is visible to anyone who finds the dead port, and the evidence is unambiguous.

5.2 Design

Port Security is applied to **every access port**, where a person can plug something in. Uplinks and core ports are excluded because they connect switches, not people. Each port learns the first MAC address it sees and keeps it (*sticky*). If another address appears, the port shuts down (decision D5).

Switch	Ports	Devices	Max MACs
SW1	Fa0/1–4	Classroom and Library PCs	1 each
SW2	Fa0/1–11	Lab PCs	1 each
SW3	Fa0/1–3	Admin PCs	1 each
SW3	Fa0/4	Printer	1
SW3	Fa0/5	Reception PC	1
SW3	Fa0/6	IP phone	2

The phone port allows two MAC addresses because the phone presents a data-side and a voice-side address. Milestone 2 problem 4 (Section 8) found that a limit of 1 err-disabled this port at start-up.

Port Security on SW3 (SW1 and SW2 use the same four lines on their access ranges)

```
interface range fastEthernet0/1 - 5
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
interface fastEthernet0/6
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
```

5.3 Verification of the configuration

After testing, each switch confirms its limits, learned addresses and action. Every access port shows zero violations.

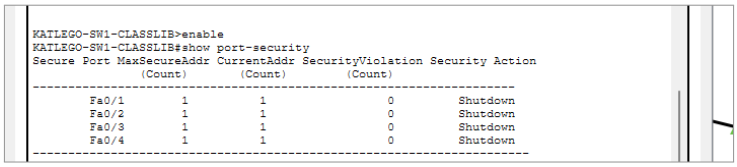


Figure 11: *SW1: Fa0/1–Fa0/4, maximum 1, action Shutdown (Milestone 2 evidence).*

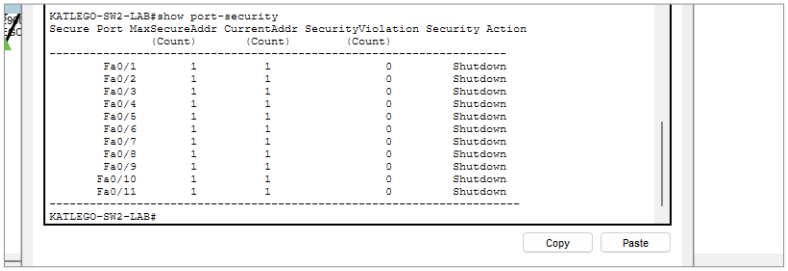


Figure 12: *SW2: Fa0/1–Fa0/11, maximum 1, action Shutdown (Milestone 2 evidence).*

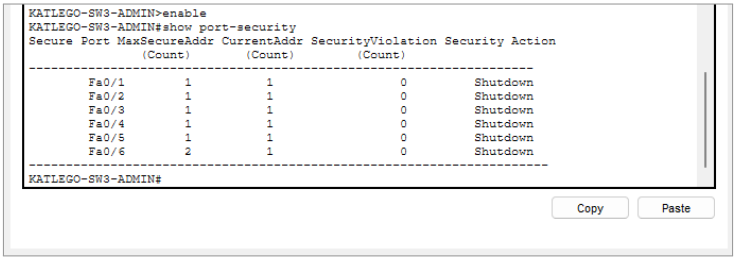


Figure 13: *SW3: Fa0/1–Fa0/5 maximum 1, phone port Fa0/6 maximum 2 (Milestone 2 evidence).*

5.4 Demonstration: a real violation (test T5)

Test T5: Port Security violation (FR4)

Aim	Show that a port shuts down when a device the switch has not learned is plugged in.
Method	On SW2 Fa0/4, PC4 was unplugged and a different PC connected and set to DHCP so that it sent traffic. Afterwards PC4 was reconnected and the port re-enabled with `shutdown` then `no shutdown`.
Expected	Violation logged, port err-disabled, other ports unaffected; after recovery PC4 is accepted again.
Result	Violation caused by MAC `0060.3E89.717E`; Fa0/4 err-disabled, Secure-shutdown, violation count 1; recovered to Secure-up with PC4's MAC `0005.5E74.0E9C` and count 0. **Pass.**

The four screenshots tell the story in order: before, violation, shutdown, recovery. PC4's MAC was never re-learned, because the sticky entry survived the shutdown.

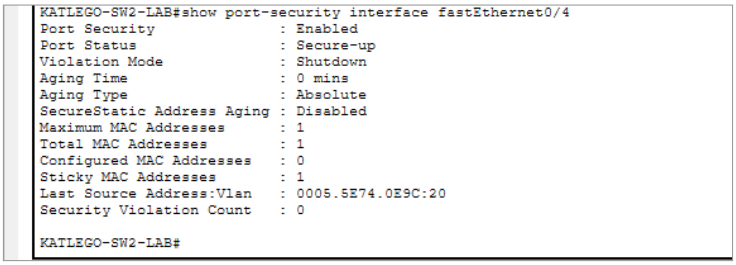


Figure 14: *Before: Fa0/4 is Secure-up with 0 violations.*

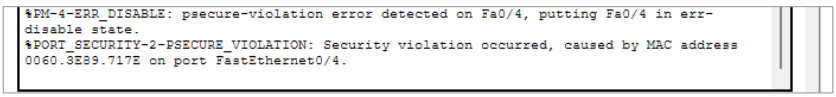


Figure 15: *The violation: the switch logs the unknown MAC and err-disables the port.*

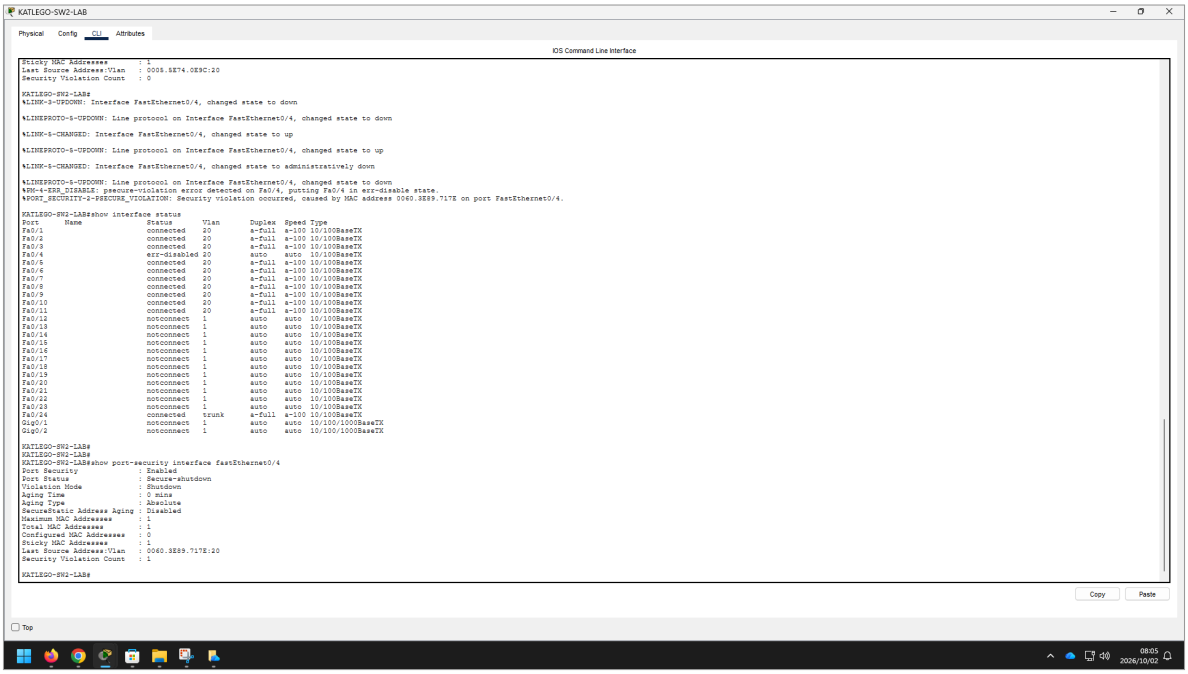


Figure 16: *The shutdown: Fa0/4 is err-disabled with violation count 1, while the other Lab ports stay connected.*

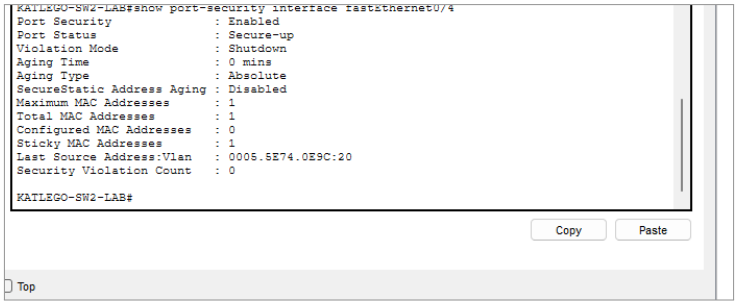


Figure 17: *The recovery: Fa0/4 is Secure-up again with PC4's MAC retained.*

5.5 Hardening: disabling unused ports (test T8)

Port Security protects ports that carry a device. Ports with nothing plugged in remained open in VLAN 1 with no protection. At the final stage every unused port was administratively shut down (decision D6), so an unauthorised device cannot connect there at all. Used ports and uplinks were not touched.

Switch	Used (unchanged)	Shut down	Count
KATLEGO-SW1-CLASSLIB	Fa0/1–4, Fa0/24	Fa0/5–23, Gig0/1–2	21
KATLEGO-SW2-LAB	Fa0/1–11, Fa0/24	Fa0/12–23, Gig0/1–2	14
KATLEGO-SW3-ADMIN	Fa0/1–6, Fa0/24	Fa0/7–23, Gig0/1–2	19
KATLEGO-CORE	Fa0/1, Fa0/2, Fa0/4, Gig0/1	Fa0/3, Fa0/5–24, Gig0/2	22
Total unused ports disabled			**76**

Unused-port shutdown (SW1 shown; the other switches use their own ranges)

```
interface range fastEthernet0/5 - 23
 shutdown
exit
interface range gigabitEthernet0/1 - 2
 shutdown
end
write memory
```

For each switch the evidence has two parts: the commands being entered and the resulting `show interfaces status`. In every status listing the used ports read `connected` and all others read `disabled`.

KATLEGO-SW1-CLASSLIB

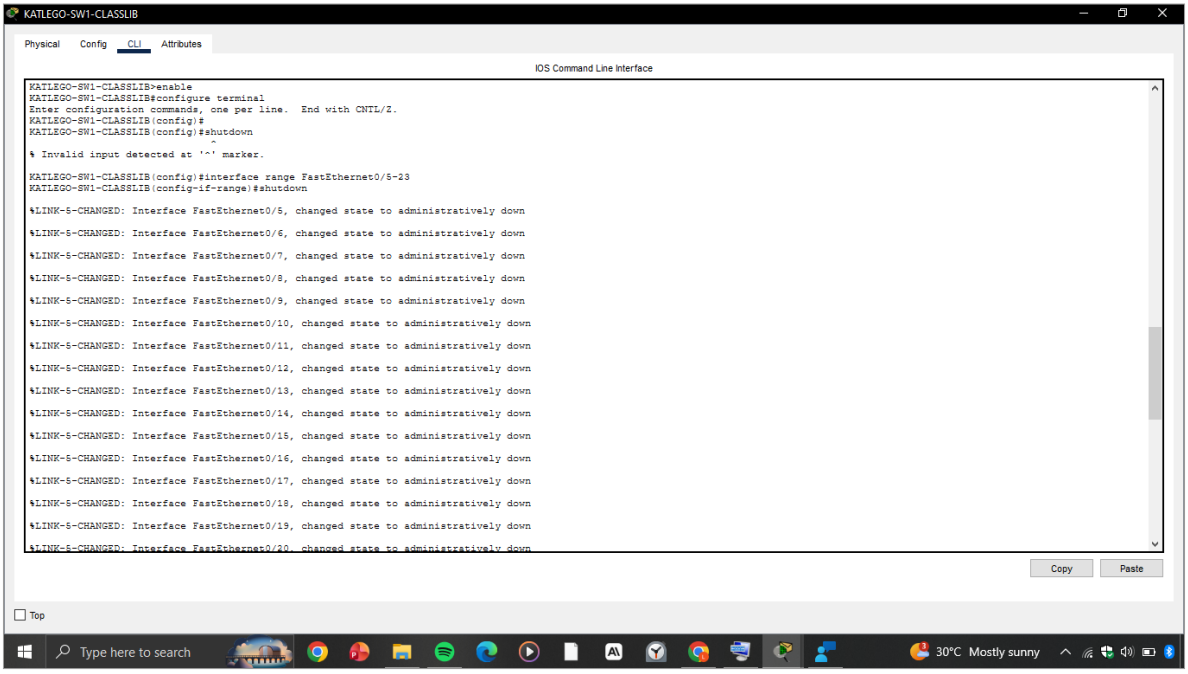


Figure 18: *SW1: unused Fast Ethernet ports shut down. The first attempt, a bare `shutdown` in global configuration mode, was rejected (`% Invalid input`); it was repeated correctly under `interface range`.*

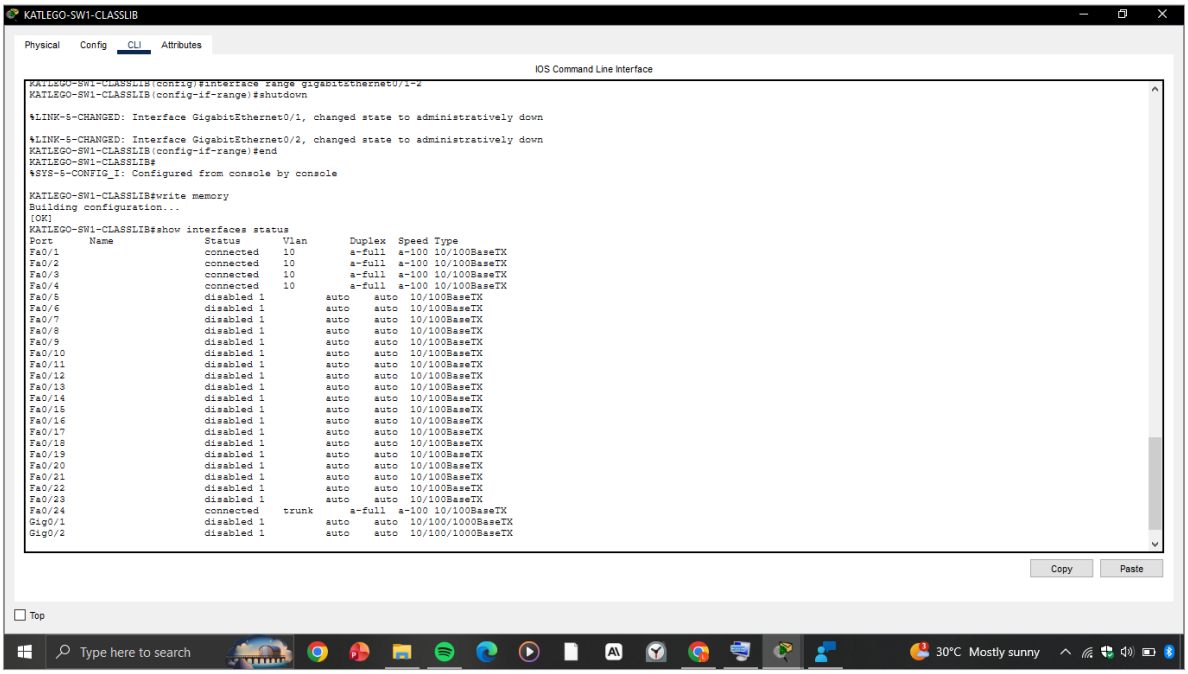


Figure 19: *SW1 `show interfaces status`: Fa0/1–4 connected in VLAN 10, Fa0/24 trunk connected, Fa0/5–23 and Gig0/1–2 disabled.*

KATLEGO-SW2-LAB

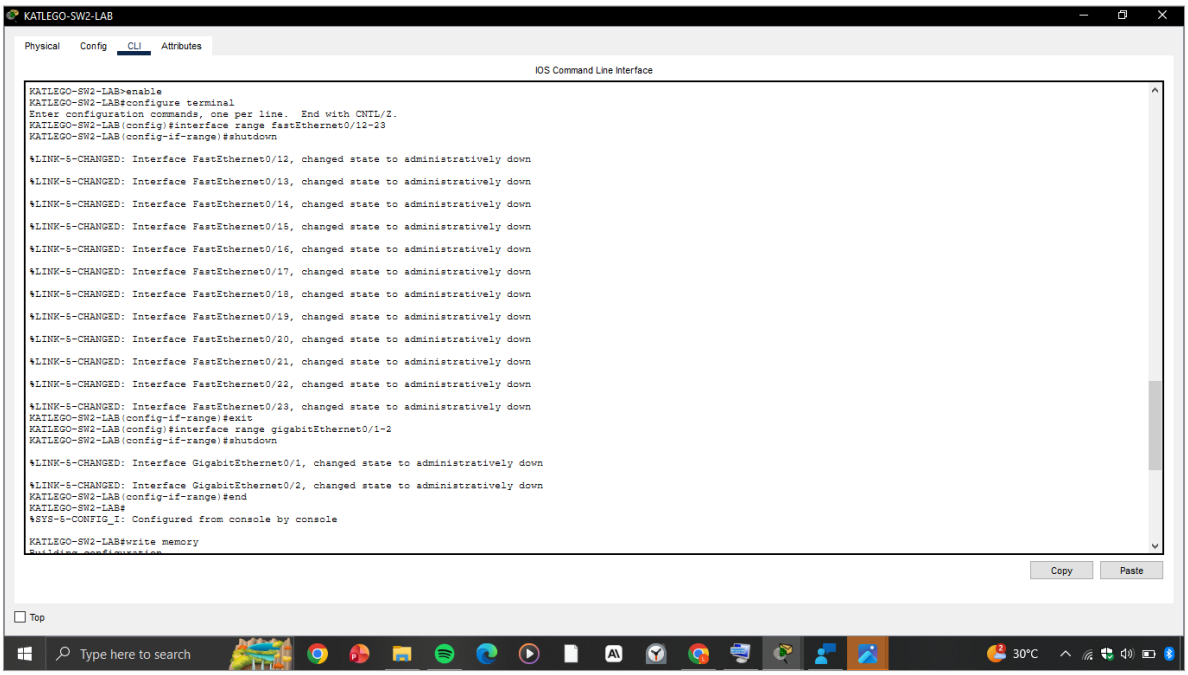


Figure 20: *SW2: Fa0/12–23 and Gig0/1–2 shut down.*

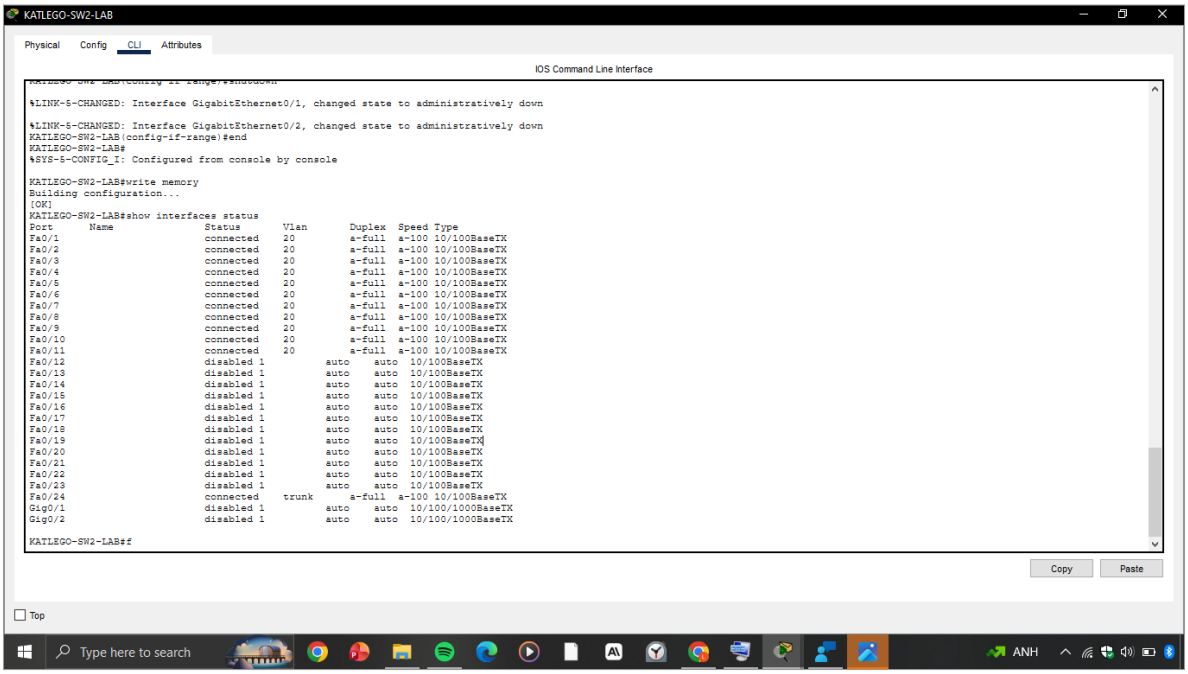


Figure 21: *SW2 `show interfaces status`: Fa0/1–11 connected in VLAN 20, Fa0/24 trunk connected, all other ports disabled.*

KATLEGO-SW3-ADMIN

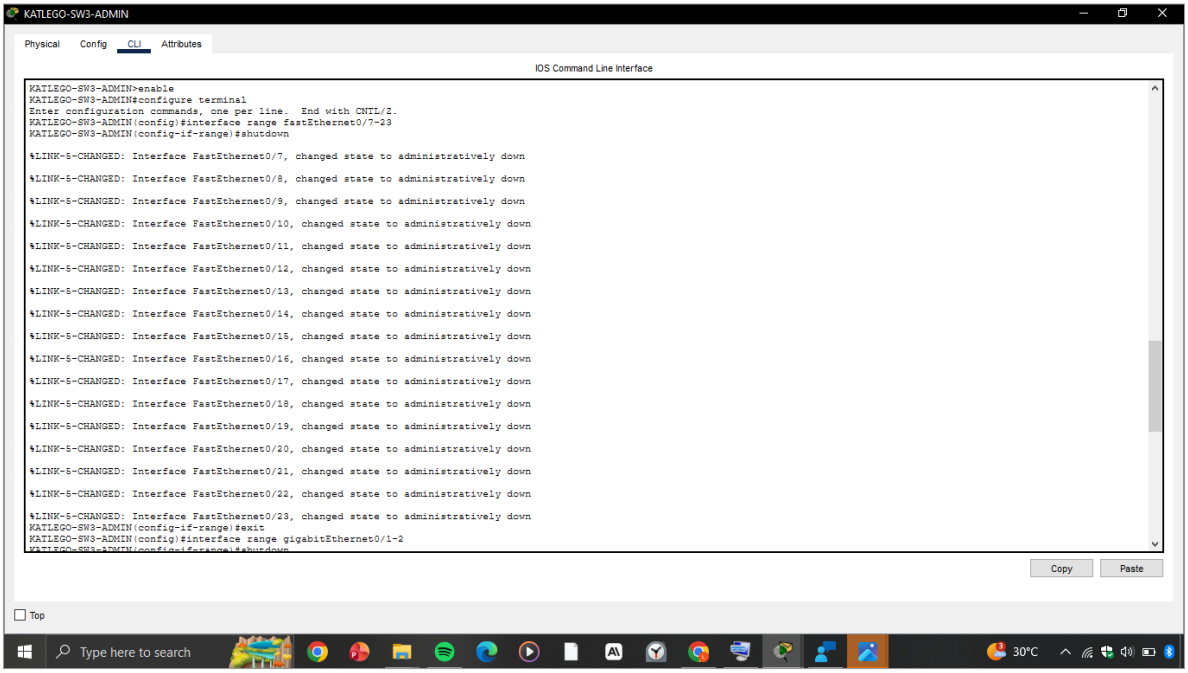


Figure 22: *SW3: Fa0/7–23 and Gig0/1–2 shut down.*

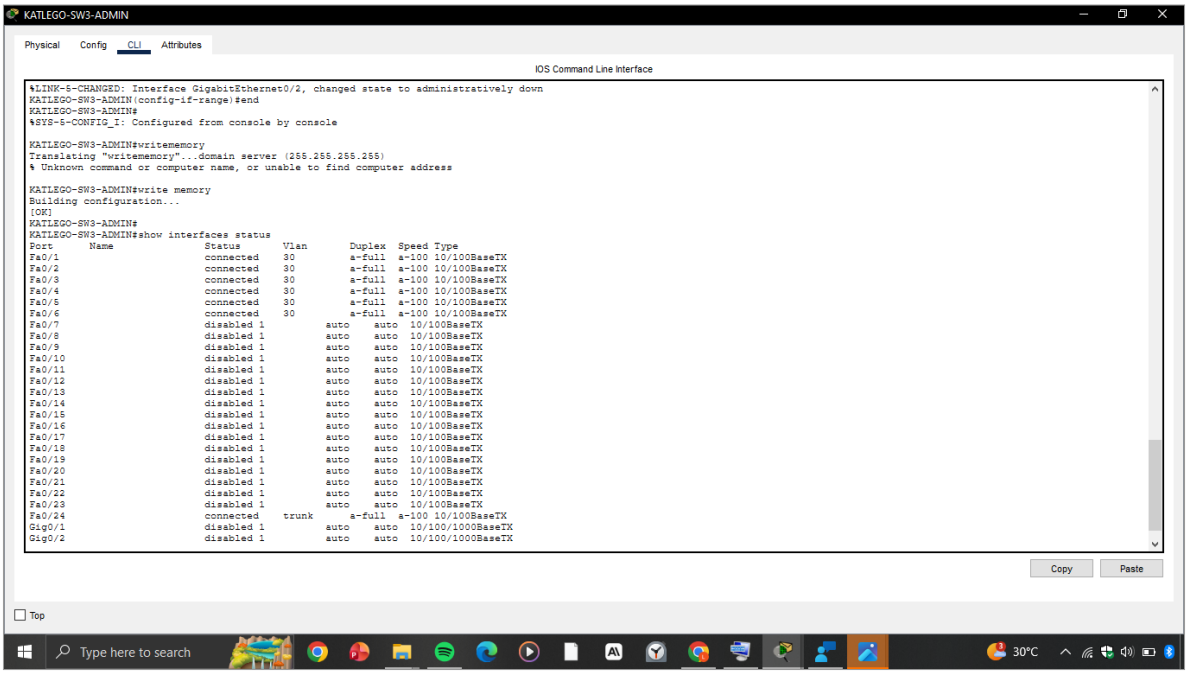


Figure 23: *SW3 `show interfaces status`: Fa0/1–6 connected in VLAN 30 (the phone port Fa0/6 stays up), Fa0/24 trunk connected, all other ports disabled. A mistyped `writememory` (no space) was rejected and re-entered as `write memory`, which returned `[OK]`.*

KATLEGO-CORE

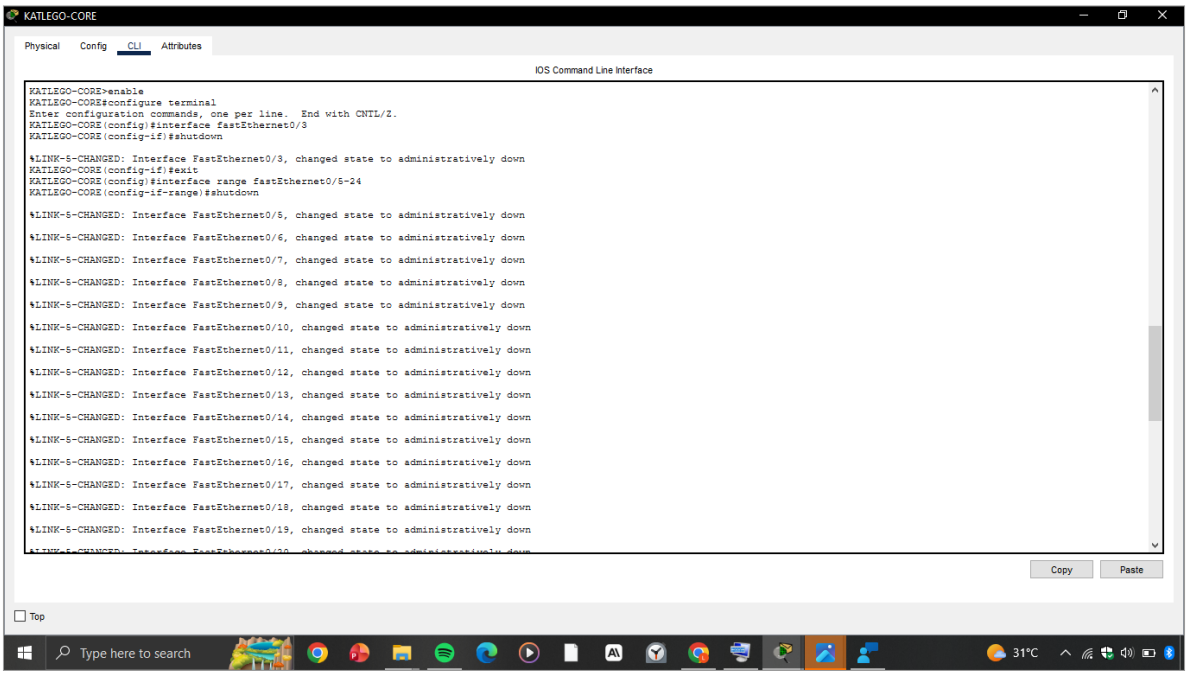


Figure 24: *Core switch: Fa0/3 and Fa0/5–24 shut down.*

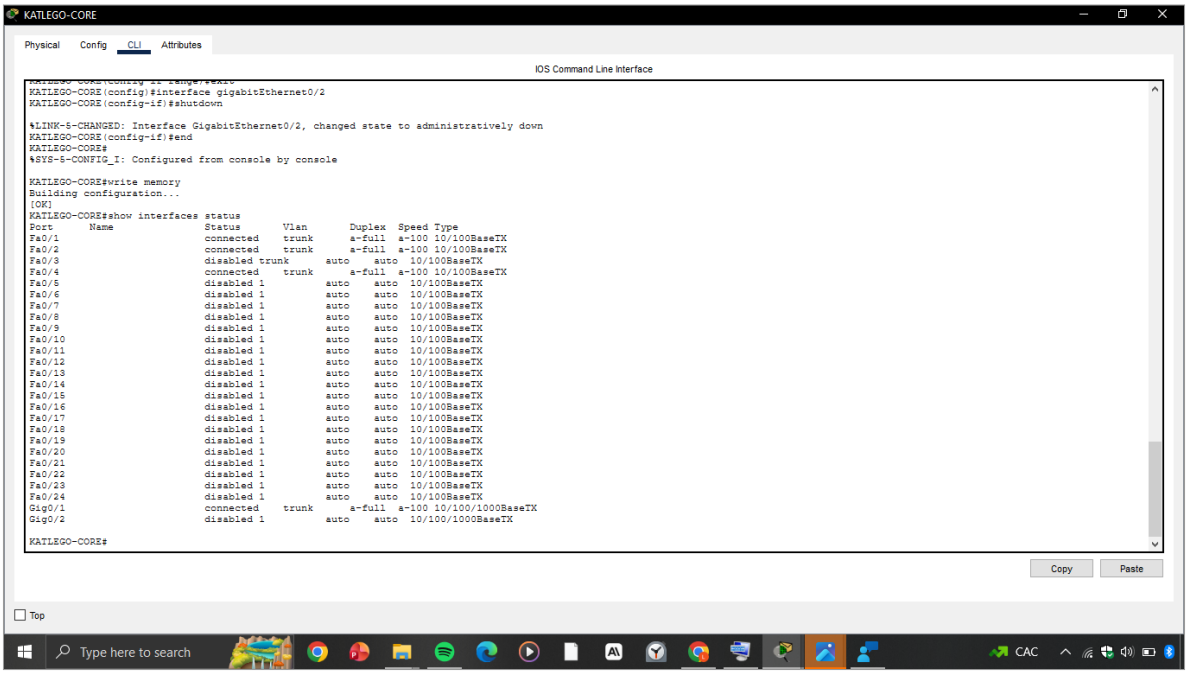


Figure 25: *Core `show interfaces status`: Fa0/1, Fa0/2, Fa0/4 and Gig0/1 connected as trunks; Fa0/3, Fa0/5–24 and Gig0/2 disabled. Fa0/3 is still configured as a trunk but is administratively down.*

5.6 Limits of this control

Port Security with sticky addresses has honest limits. A MAC address can be spoofed, so it identifies a device weakly; the first device to plug into an empty secured port is trusted by default; and an err-disabled port needs a manual `shutdown`/`no shutdown` to recover, which is acceptable for a centre where a visibly dead port prompts action but would be an overhead at larger scale. Stronger controls such as 802.1X are discussed in Section 9.

6 Protecting the server room

Milestone 1 gave the server room its own subnet (VLAN 99, `192.168.22.72/29`). But once routing is enabled every VLAN could still reach it, so a subnet alone does not satisfy FR5. An access list now blocks VLANs 10, 20 and 40 from that subnet (decision D7).

KATLEGO-RTR: ACL 100

```
access-list 100 remark Deny non-Admin VLANs from reaching the Server Room segment
access-list 100 deny ip any 192.168.22.72 0.0.0.7
access-list 100 permit ip any any
interface gigabitEthernet0/0.10
 ip access-group 100 in
interface gigabitEthernet0/0.20
 ip access-group 100 in
interface gigabitEthernet0/0.40
 ip access-group 100 in
```

Wildcard mask for the protected subnet

1. The protected subnet is `192.168.22.72/29`, mask 255.255.255.248.
2. The wildcard is the inverse of the mask: $255 - 248 = 7$ in the last octet, so 0.0.0.7.
3. The wildcard matches 8 addresses, from `.72` to `.79`: exactly the VLAN 99 block.

Answer: **192.168.22.72 0.0.0.7**

Design notes. The list is applied **inbound on the source VLAN sub-interfaces**, so blocked packets are dropped as early as possible. The final `permit ip any any` is essential: every access list ends with an invisible deny, which would otherwise block all routing. The list is not applied to VLAN 30, the deliberate exception (assumption 1), or to VLAN 99 itself. Traffic from the restricted VLANs to the Internet still passes, because only the server-room block is denied.

Test T3: Restricted access to the server room (FR5)

Aim	Show that Classroom and Lab devices cannot reach the server-room segment.
Method	PC0 (VLAN 10) and PC4 (VLAN 20) each pinged the core switch at 192.168.22.74. Then `show access-lists 100` was run on the router.
Expected	Both fail; the `deny` line counts the blocked packets.
Result	Destination host unreachable from the router for both; the `deny` entry shows 4 matches. **Pass.**

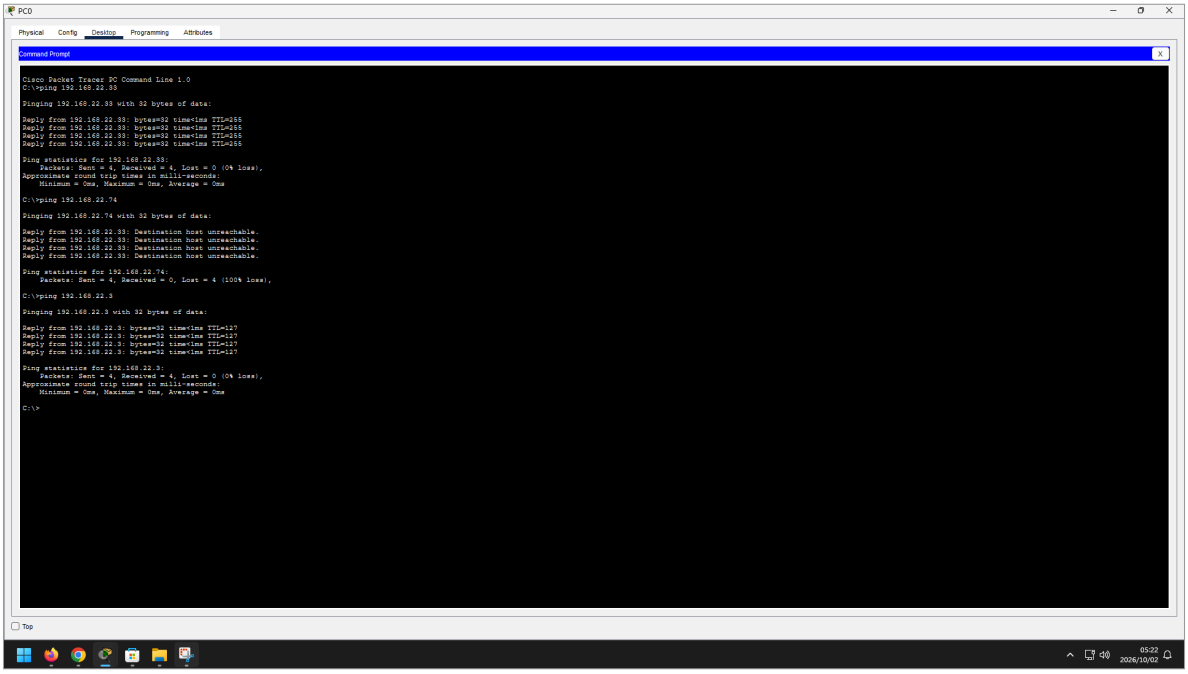


Figure 26: *Classroom PC0 is blocked from the server-room segment (Milestone 2 evidence).*

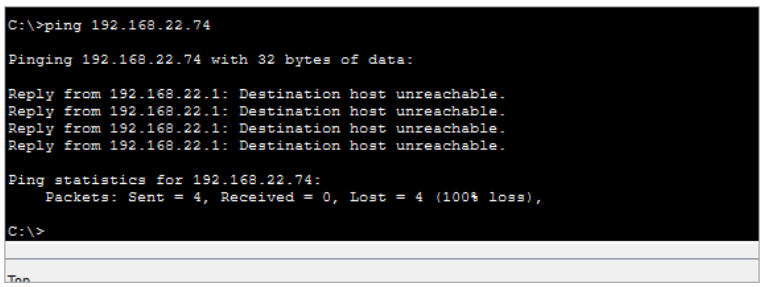


Figure 27: *Lab PC4 is blocked from the server-room segment (Milestone 2 evidence).*

```
KATLEGO-RTR#
KATLEGO-RTR#show access-lists 100
Extended IP access list 100
    deny ip any 192.168.22.72 0.0.0.7 (4 match(es))
    permit ip any any (46 match(es))

KATLEGO-RTR#
```

Figure 28: *ACL 100 on the router: the `deny` entry counts 4 matches (Milestone 2 evidence).*

Test T4: The Admin exception

Aim	Show that Admin, and only Admin, can reach the server room.
Method	PC7 (VLAN 30) pinged 192.168.22.74.
Expected	Replies with TTL 254: the packets crossed the router and were allowed through.
Result	4 of 4 replies, 0% loss, TTL 254. **Pass.**

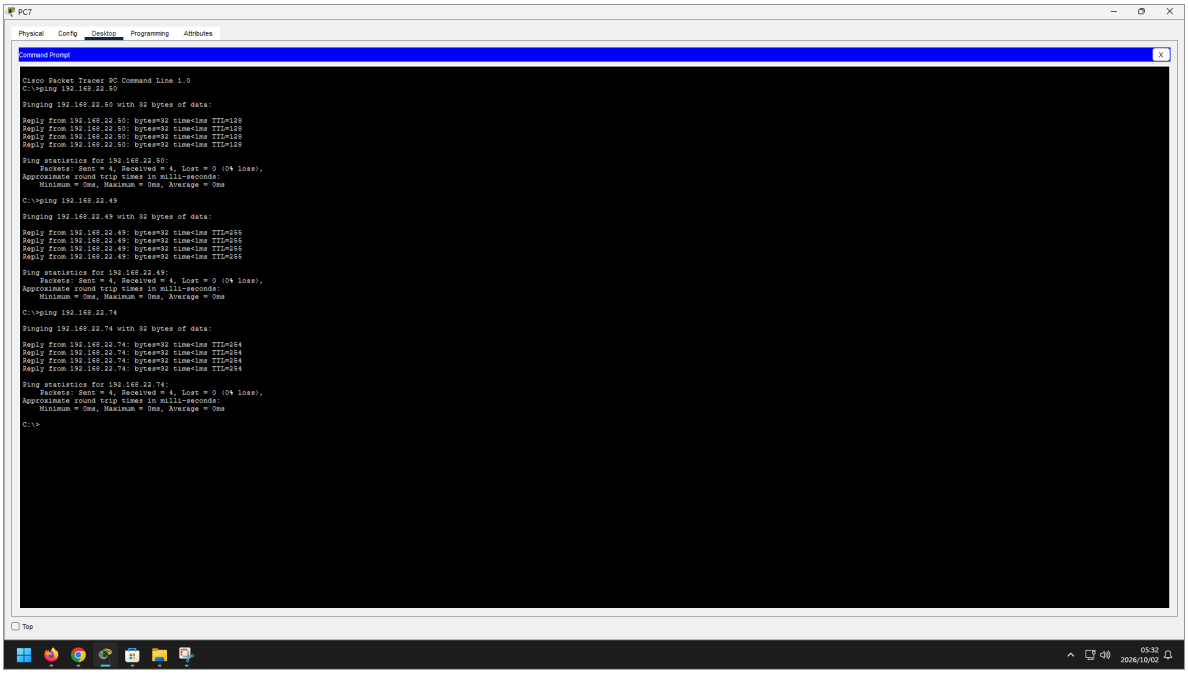


Figure 29: *Admin PC7 reaches the server-room segment (Milestone 2 evidence).*

7 Testing and verification

7.1 Strategy

Every claim in this report is tested. Tests follow five questions: **reach** (can devices talk where they should?), **restrict** (who is kept out?), **react** (what happens when an unknown device appears?), **serve** (do devices get what they need?) and **connect** (can the network reach the outside and resolve names?). After the final-stage changes, three **regression tests** confirmed that earlier results still hold.

7.2 Results at a glance

ID	Test	Observed	Result
T1	Lab PC to Lab PC (same VLAN)	4 of 4 replies, TTL 128 (no router hop)	Pass
T2	Classroom PC to Lab PC (across VLANs)	4 of 4 replies, TTL 127 (one router hop)	Pass

ID	Test	Observed	Result
T3	Classroom and Lab PC to server room	Destination host unreachable; ACL deny counts 4 matches	Pass
T4	Admin PC to server room	4 of 4 replies, TTL 254	Pass
T5	Unknown device on a secured port	Port err-disabled, violation counted, then recovered	Pass
T6	DHCP leasing	20 leases, all in range, none excluded	Pass
T7	Admin PC to IP phone	4 of 4 replies, TTL 254	Pass
T8	Unused ports disabled (all four switches)	Used ports `connected`; every unused port `disabled`; 76 ports in total	Pass
T9	Internet edge: interface and default route	Gi0/1 203.0.113.2 up/up; `S*` default route via 203.0.113.1	Pass
T10	Router to ISP and DNS server	203.0.113.1: 5/5; 8.8.8.8: first run 4/5 (ARP), repeat 5/5	Pass
T11	Lab PC (VLAN 20) to 8.8.8.8	4 of 4 replies, TTL 126, average 14 ms	Pass
T12	Classroom PC (VLAN 10) to 8.8.8.8	4 of 4 replies, TTL 126, average 19 ms	Pass
T13	Admin PC (VLAN 30) to 8.8.8.8	4 of 4 replies, TTL 126, average 16 ms	Pass
T14	DNS lookup and ping by name	`nslookup www.katlego.edu` returns 8.8.8.8; ping by name 4 of 4, TTL 126	Pass
T15	Regression: Classroom to server room	Still blocked: 4 × destination host unreachable, 100% loss	Pass
T16	Regression: Admin to server room	First run 2 of 4 (ARP warm-up), repeat run 4 of 4, TTL 254	Pass
T17	Regression: Port Security on SW2	Fa0/1–11: max 1, current 1, 0 violations, action Shutdown	Pass

7.3 Reach: can devices talk to each other?

Test T1: Connectivity within a VLAN

Aim	Show that devices in the same VLAN communicate directly.
Method	PC4 (192.168.22.3) pinged another Lab PC, 192.168.22.2.
Expected	Replies with TTL 128: the packets never crossed the router.
Result	4 of 4 replies, 0% loss, TTL 128. **Pass.**

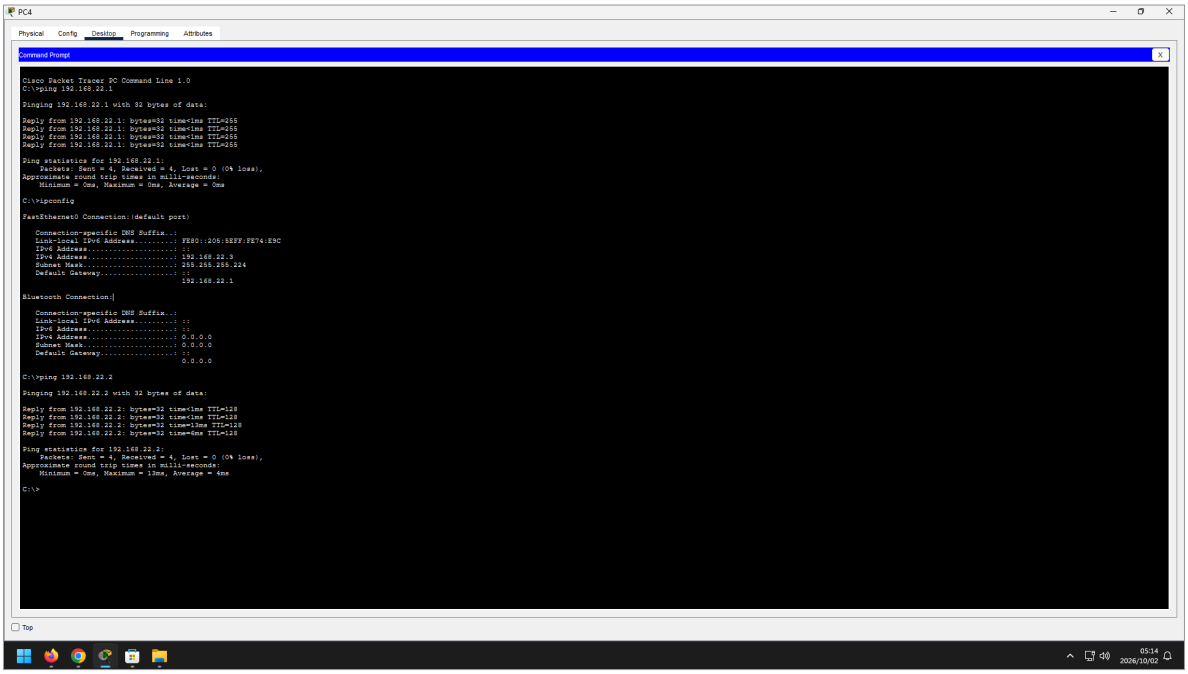


Figure 30: *PC4 reaches its gateway and another Lab PC (Milestone 2 evidence).*

Test T2: Connectivity between VLANs

Aim	Show that the router joins the VLANs.
Method	PC0 (VLAN 10) pinged PC4 (192.168.22.3, VLAN 20).
Expected	Replies with TTL 127: one router hop took 1 off 128.
Result	4 of 4 replies, 0% loss, TTL 127. **Pass.**

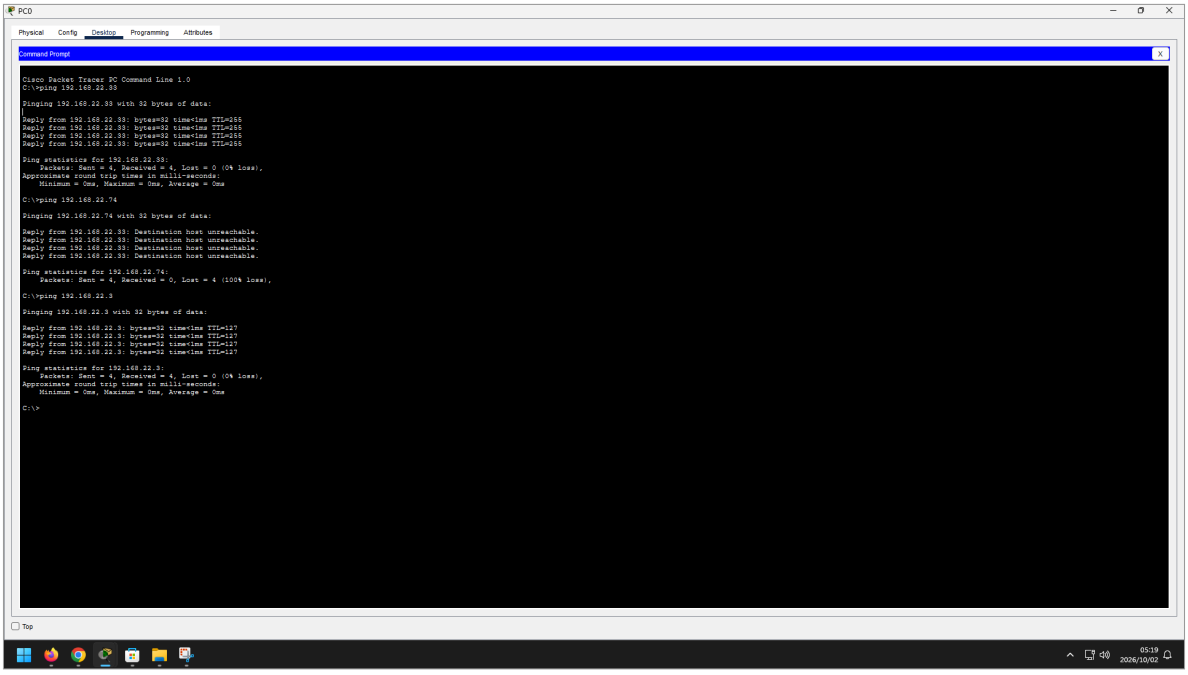


Figure 31: *A Classroom PC reaches a Lab PC through the router (Milestone 2 evidence).*

Tests T3 and T4 (server-room restriction) are in Section 6, T5 (Port Security violation) and T8 (unused ports) in Section 5, and T6 (DHCP) and T7 (VoIP) in Section 4.

7.4 Connect: the Internet edge (final stage)

Tests T9 and T10: Edge link and router reachability

Aim	Show that the router's edge interface is up, the default route is installed, and the ISP router and DNS server answer.
Method	On KATLEGO-RTR: `show ip interface brief`, `show ip route`, `ping 203.0.113.1`, `ping 8.8.8.8`.
Expected	Gi0/1 up/up with 203.0.113.2; an `S*` route via 203.0.113.1; pings succeed.
Result	As expected (Figure 10). `ping 203.0.113.1`: `!!!!!`, 5/5. `ping 8.8.8.8`: first run `.!!!!` (4/5, 80%) because the first packet waits for address resolution, repeat run `!!!!!` (5/5, 100%). **Pass.**

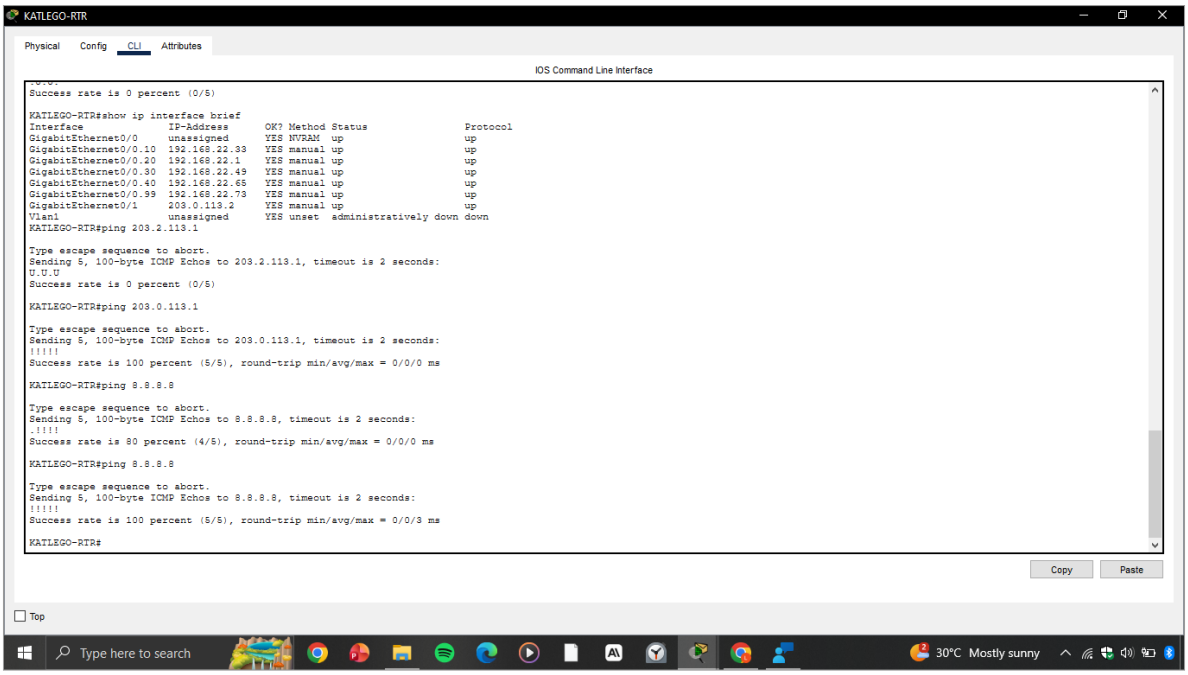


Figure 32: *KATLEGO-RTR pings the ISP router (5/5) and the DNS server (80% on the first run, 100% on the repeat). The two `U` results at the top came from a mistyped test address and are explained in Section 8.*

Tests T11 to T13: every user VLAN reaches the Internet

Aim	Show that devices in three different VLANs reach 8.8.8.8 through both routers.
Method	`ping 8.8.8.8` from PC4 (Lab, VLAN 20), PC0 (Classroom, VLAN 10) and PC7 (Admin, VLAN 30).
Expected	4 replies, TTL 126: a PC starts at 128 and the reply crosses two routers.
Result	All three: 4 of 4 replies, 0% loss, TTL 126. **Pass.**

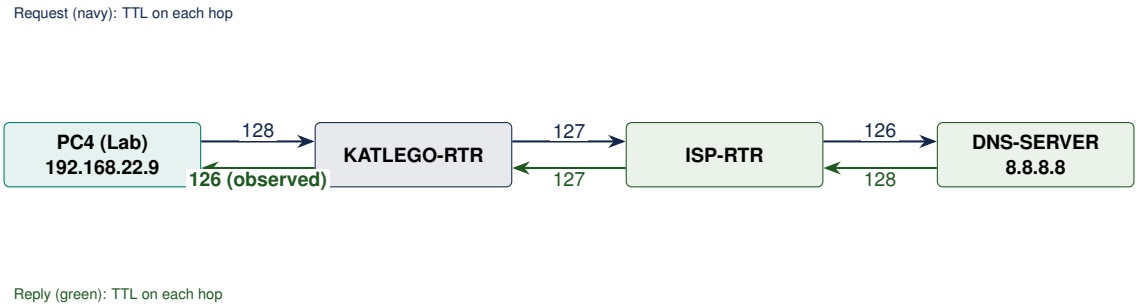


Figure 33: *Why a reply from 8.8.8.8 shows TTL 126: the reply starts at 128 and each of the two routers subtracts 1.*

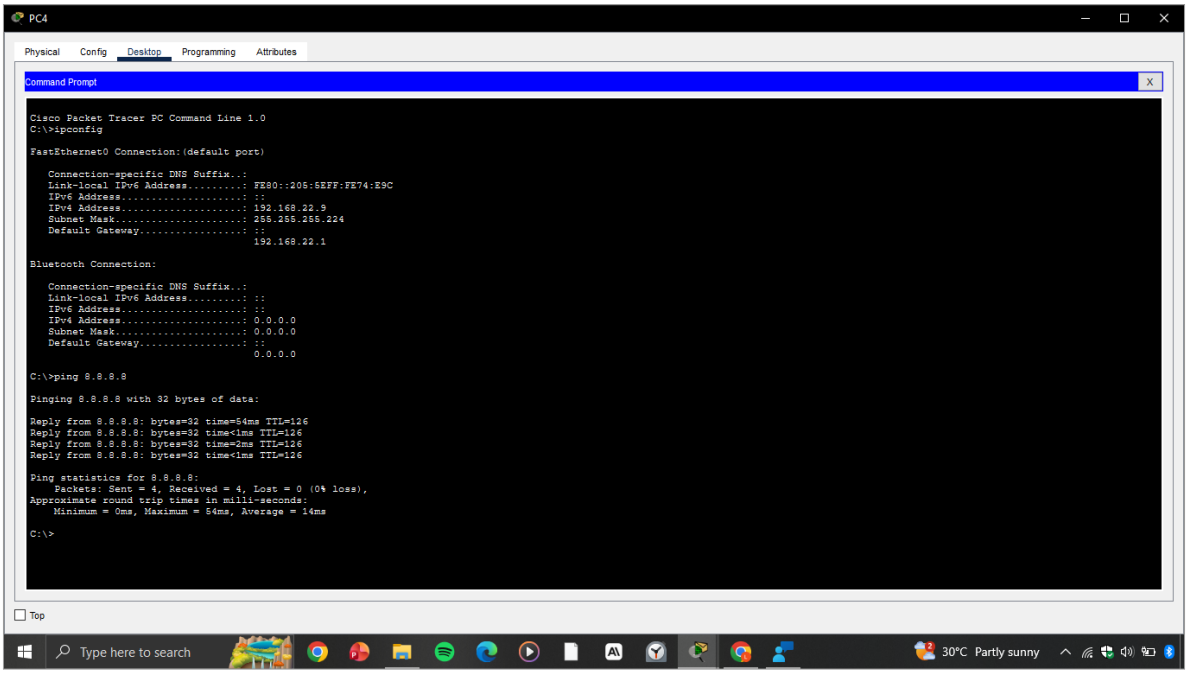


Figure 34: *T11: Lab PC4 (192.168.22.9, VLAN 20) pings 8.8.8.8: 4 of 4 replies, TTL 126.*

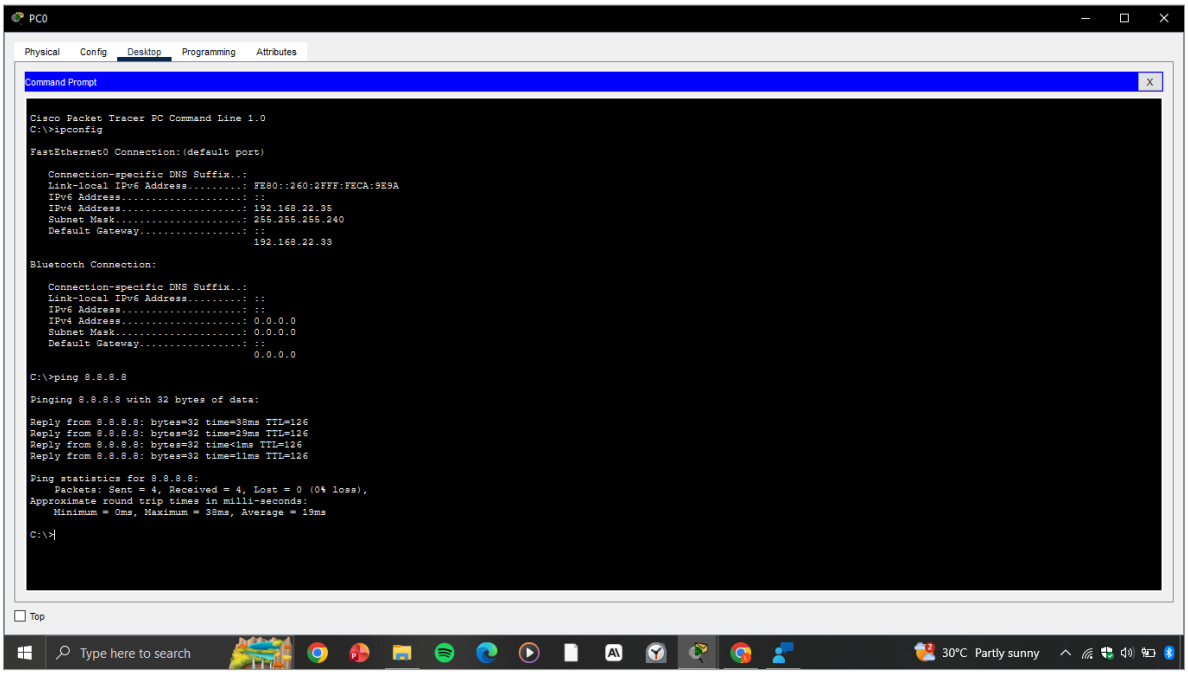


Figure 35: *T12: Classroom PC0 (192.168.22.35, VLAN 10) pings 8.8.8.8: 4 of 4 replies, TTL 126.*

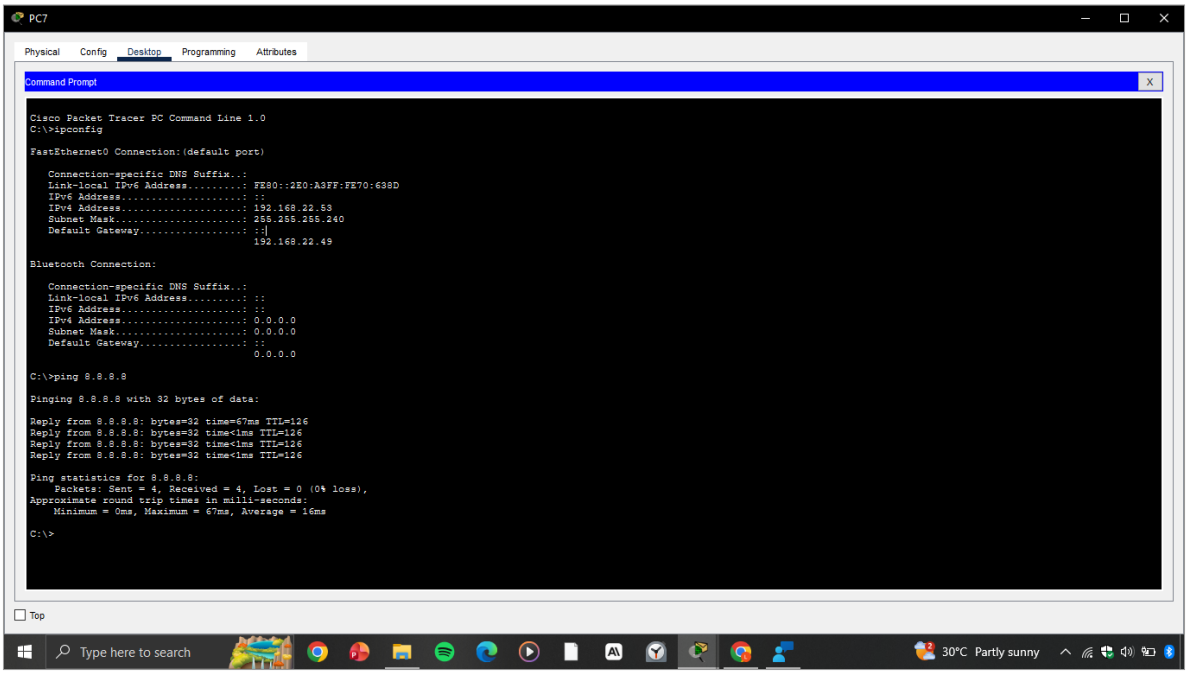


Figure 36: *T13: Admin PC7 (192.168.22.53, VLAN 30) pings 8.8.8.8: 4 of 4 replies, TTL 126.*

Test T14: DNS name resolution	
Aim	Show that a PC asks the DNS server for the address of a name and receives the answer.
Method	On PC0: `nslookup www.katlego.edu`, then `ping www.katlego.edu`.
Expected	The name resolves to 8.8.8.8 and the ping goes to that address.
Result	`nslookup` reports Server 8.8.8.8 and a non-authoritative answer, name `www.katlego.edu`, address 8.8.8.8. The ping by name returns 4 of 4 replies, TTL 126. **Pass.**

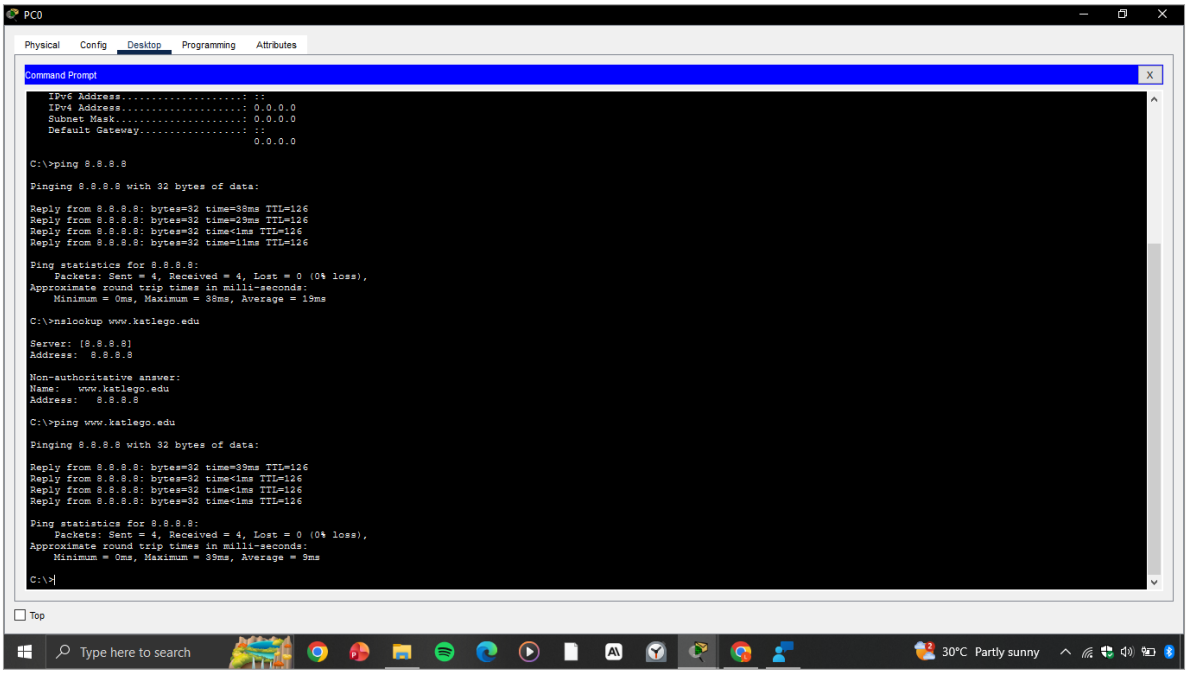


Figure 37: *T14: PC0 resolves `www.katlego.edu` through the DNS server and pings it by name.*

7.5 Regression: did the final changes break anything?

A change can break something that used to work. After the unused ports were shut and the Internet edge added, three earlier results were re-tested.

Tests T15 and T16: server-room restriction still holds

Method	PC0 (Classroom) and PC7 (Admin) each pinged 192.168.22.74.
Expected	Classroom blocked by ACL 100; Admin reaches the server room with TTL 254.
Result	PC0: 4 × `Destination host unreachable` from 192.168.22.33, 100% loss. PC7: first run 2 of 4 replies (the first two requests timed out while addresses were being resolved after the topology change), repeat run 4 of 4, TTL 254, 0% loss. **Pass.**

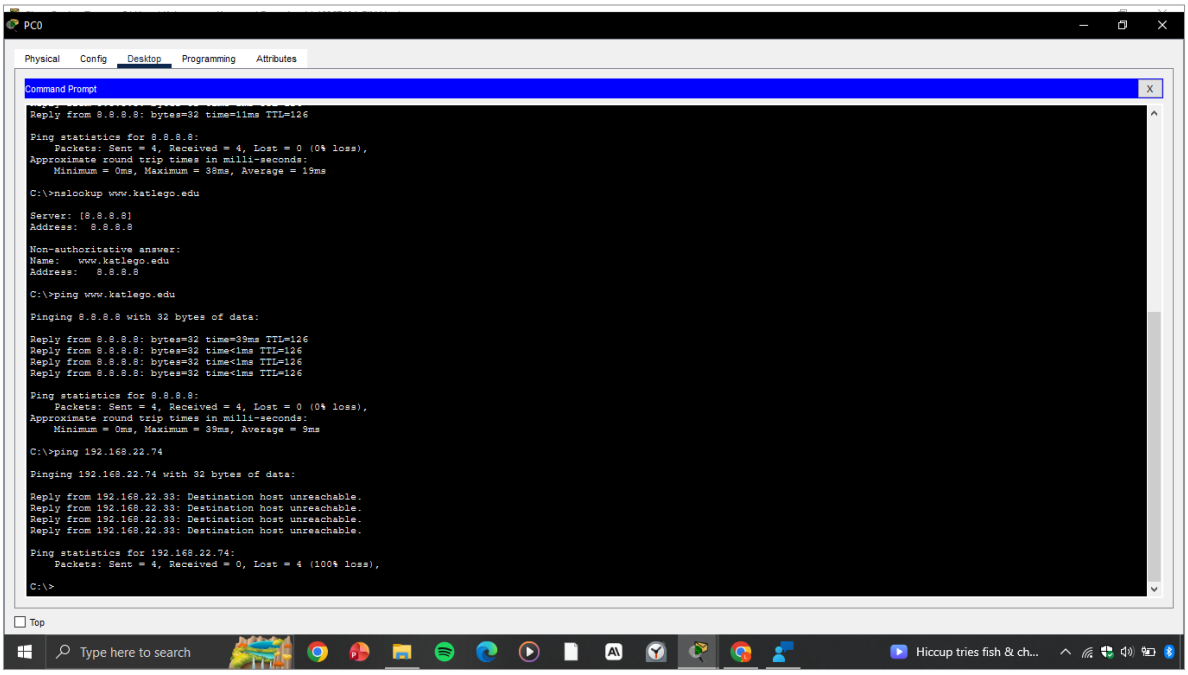


Figure 38: *T15: PC0 (Classroom) is still blocked from 192.168.22.74. The upper part of the screen repeats the DNS test.*

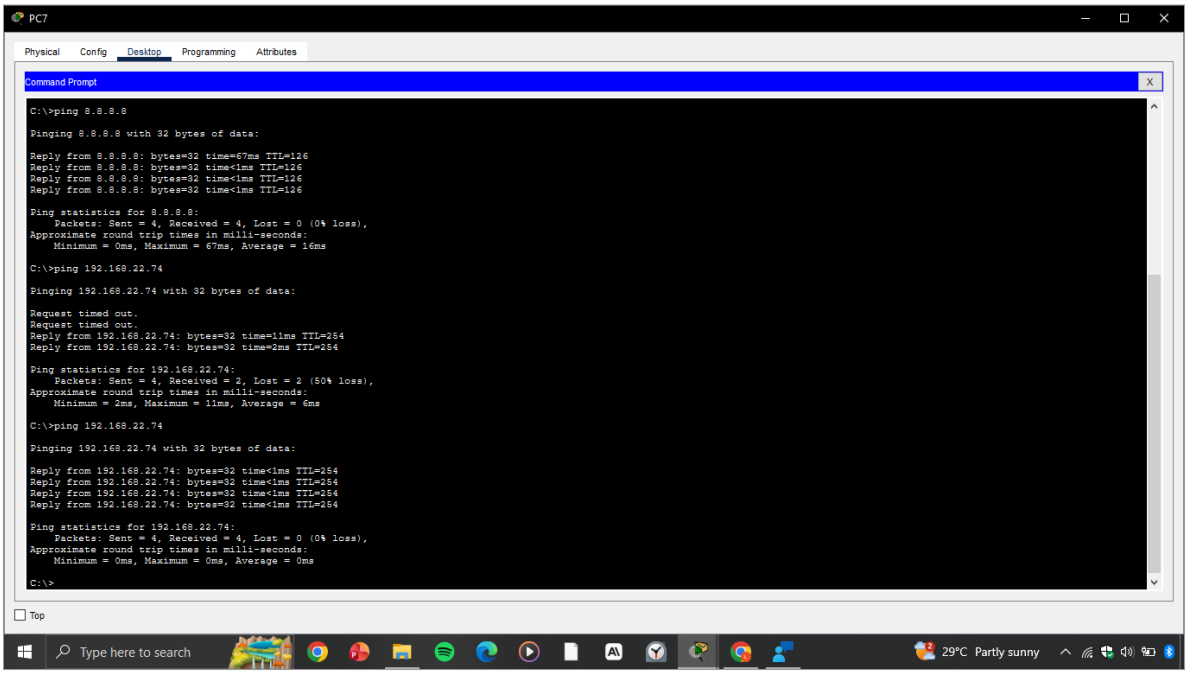


Figure 39: *T16: PC7 (Admin) still reaches 192.168.22.74. The first run loses 2 of 4 packets while addresses are resolved; the repeat run is 4 of 4, TTL 254.*

Test T17: Port Security still healthy

Method	`show port-security` on SW2.
Expected	Fa0/1–Fa0/11 each: maximum 1, current 1, 0 violations, action Shutdown, matching Milestone 2.
Result	As expected. **Pass.**

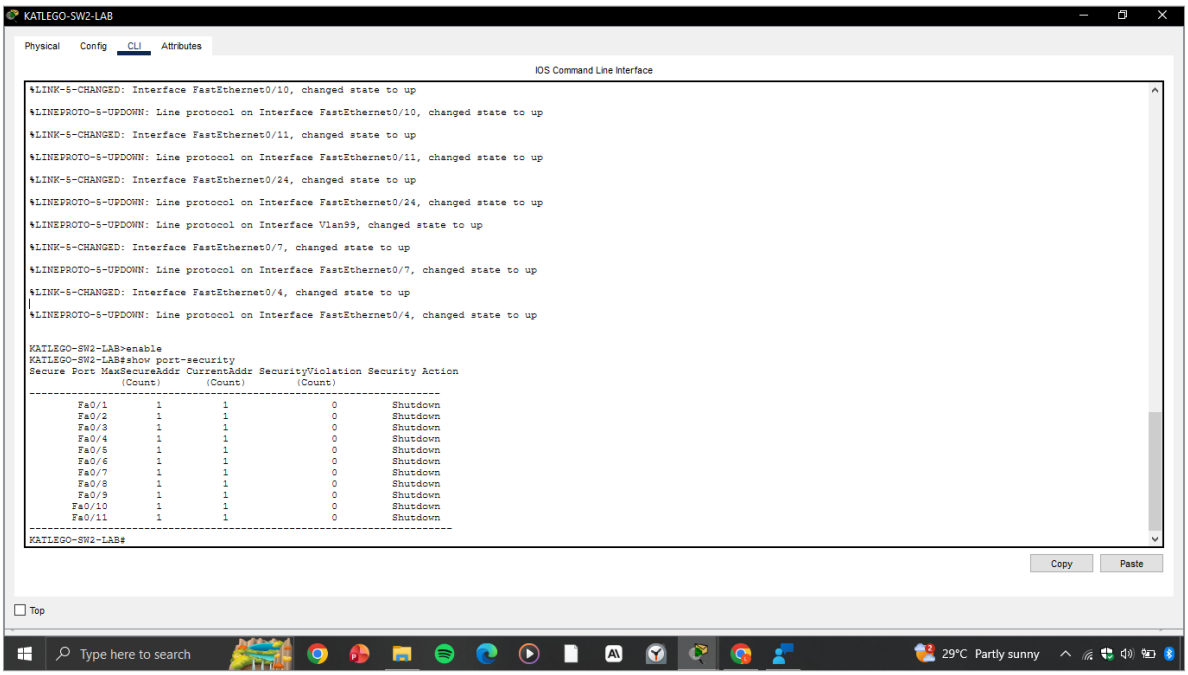


Figure 40: *T17: `show port-security` on SW2 after the final changes: all eleven access ports show maximum 1, current 1, violation count 0, action Shutdown.*

7.6 Requirement traceability

ID	Requirement	Implemented by	Verified by
FR1	Connectivity	VLANs, router-on-a-stick, trunks	T1, T2, T11–T13
FR2	Segmentation	Five VLANs, pruned trunks	T2, T3, T4
FR3	Scalable addressing	VLSM plan, DHCP	T6; Section 3
FR4	Port-level access control	Sticky Port Security; unused ports shut	T5, T8, T17
FR5	Restricted infrastructure	VLAN 99 and ACL 100	T3, T4, T15, T16
FR6	Voice readiness (CR11)	VLAN 40, voice VLAN, option 150	T7
FR7	External access and DNS	ISP router, default route, DNS server	T9–T14

8 Problems found and how they were fixed

The first build did not pass every test. Problems were found by testing and fixed at their cause. Problems 1–6 arose during Milestone 2; problems 7–10 during the final stage.

No.	What was seen	Cause	Fix
1	DHCP address conflict logged for 192.168.22.33	Exclusion range typed with its end below its start, leaving the VLAN 10 gateway unprotected	Removed it and added a single `excluded-address 192.168.22.33`
2	Voice pool incomplete	Option 150 missing	Added `option 150 ip 192.168.22.65`
3	SW3 trunk on the core carrying every VLAN	Core port left to negotiate, so it was unpruned	Fixed trunk allowing only VLANs 30, 40, 99
4	Phone port err-disabled at start-up	Two different MAC addresses seen on the port, with a maximum of 1	Maximum set to 2 on that port
5	Phone port blocked by spanning tree; no ping replies	Port was a plain VLAN 40 access port with no voice VLAN	Access VLAN 30 with `switchport voice vlan 40`
6	ACL missing from the voice VLAN	Applied only to VLANs 10 and 20	Applied to Gi0/0.40
7	`% Invalid input detected` on KATLEGO-RTR when adding the default route and edge address	Typing errors: dots instead of spaces in `0.0.0.0 0.0.0.0`, an extra octet in the address, and `ip address` typed in global mode instead of under the interface	Re-entered `ip route 0.0.0.0 0.0.0.0 203.0.113.1` and the address under `interface gigabitEthernet0/1`; confirmed with `show ip route`
8	Test pings to `203.0.223.2` and `203.2.113.1` returned `U` (unreachable)	Wrong test addresses, not a network fault; the router correctly had no path	Pinged the correct neighbour, 203.0.113.1, which succeeded 5/5

No.	What was seen	Cause	Fix
9	First ping after reconfiguration lost a packet (router to 8.8.8.8: 4/5; Admin to server room: 2/4)	Normal address-resolution delay when devices first learn each other	Repeated the test; 5/5 and 4/4
10	Bare `shutdown` rejected on SW1; `writememory` rejected on SW3	`shutdown` typed in global mode instead of under `interface range`; missing space in `write memory`	Re-entered correctly; verified with `show interfaces status` and the `[OK]` from `write memory`

Lesson. In every case the error was found by reading the device's own output, not by assuming. The practice used throughout is to type a change, then prove it with a `show` command or a test before moving on.

9 Evaluation, limitations and future improvements

9.1 What worked

The design is small enough to operate without IT staff and still demonstrates sound practice: functional VLANs, VLSM with headroom, pruned trunks, secure-by-default access ports, a protected infrastructure subnet, and a clear path to the outside. Every requirement traces to a decision and every decision to a test (Section 7).

9.2 Limitations and how to address them

Limitation	Detail	Improvement
Single points of failure	One router, one trunk to the core and a single core switch; the router-on-a-stick link also limits inter-VLAN throughput.	Layer 3 core switch; redundant links and a second router with first-hop redundancy
No address translation	Private addresses reach the simulated Internet only because the ISP holds a static return route. A real provider would not route 192.168.22.0/24.	Configure NAT/PAT on KATLEGO-RTR
Port Security strength	MAC addresses can be spoofed; the first device on an empty port is trusted; recovery is manual.	802.1X, DHCP snooping, dynamic ARP inspection, `errdisable recovery`
Blunt server-room ACL	Denies all traffic from VLANs 10, 20, 40 to the server-room subnet, and trusts Admin completely.	Named extended ACL permitting only specific management protocols from named hosts, with logging
Management-plane hardening	The documented configurations do not include encrypted privileged-mode passwords, SSH or logging.	Add secrets, SSH-only management and syslog
VLAN 99 headroom	Five of six usable addresses are in use (one spare, .78).	Use the reserved block from .80 onward for a larger management subnet
Voice quality	One handset and no QoS configured.	QoS marking and trust on voice ports before wider rollout

9.3 Reflection

The most valuable lesson was that a design on paper is only a hypothesis until it has been built and tested: six problems in Milestone 2 and four more in the final stage were invisible in the design and obvious in the test output. The second lesson was scope discipline: the client is small, so every addition, from the single voice handset to the single default route, was sized to the problem rather than to the technology available. The third was that security is layered: the subnet separates, the ACL restricts, Port Security guards the edge, and disabling unused ports removes the last open doors.

10 Portfolio of evidence and submission contents

The project is assessed through three compulsory evidence streams, which together with this report make up the final submission.

No.	Deliverable	Content
1	Packet Tracer project (`.pkt`)	Fully working simulation: five VLANs, router-on-a-stick, DHCP, voice, Port Security, ACL 100, unused ports disabled, ISP router and DNS server
2	GitHub portfolio	`https://github.com/konyanalethabo-work/katlego-network-project`: the evidence record of planning, design, implementation, testing and reflection, including the milestone documents and the final-stage screenshots used in this report
3	Technical report	This document
4	Inset video demonstration (15–20 minutes)	Walk-through of the design decisions (Sections 3, 4), the assigned feature (Section 5) and the verification tests (Section 7), with the presenter visible throughout
5	Additional files	Milestone 1 and Milestone 2 documents, as required

11 Conclusion

The Katlego Adult Learning Centre network meets all seven functional requirements. Five VLANs keep classrooms, the lab, administration, voice and infrastructure apart; VLSM uses 80 of the 256 available addresses and holds 176 in reserve; Port Security guards all 21 access ports and was demonstrated by provoking and recovering from a real violation; 76 unused ports are switched off; the server-room subnet is protected by ACL 100 with a documented Admin exception; and a simulated ISP with a DNS server gives every user VLAN Internet reach and name resolution. Seventeen tests were run and all passed, including three regression tests after the final changes. The limitations in Section 9 are stated openly, together with the improvements a production deployment would add.

A Device configurations

The listings below are the essential configuration of each device as built. Lines the device adds automatically are omitted.

KATLEGO-CORE

```
hostname KATLEGO-CORE
vlan 10
 name DATA_CLASSROOMS_LIBRARY
vlan 20
 name DATA_COMPUTER_LAB
vlan 30
 name DATA_ADMIN
vlan 40
 name VOICE
vlan 99
 name MANAGEMENT
interface vlan 99
 ip address 192.168.22.74 255.255.255.248
 no shutdown
ip default-gateway 192.168.22.73
interface fastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99
interface fastEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 20,99
interface fastEthernet0/4
 switchport mode trunk
 switchport trunk allowed vlan 30,40,99
interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,99
! unused ports (final stage)
interface fastEthernet0/3
 shutdown
interface range fastEthernet0/5 - 24
 shutdown
interface gigabitEthernet0/2
 shutdown
```

KATLEGO-SW1-CLASSLIB

```
hostname KATLEGO-SW1-CLASSLIB
vlan 10
 name DATA_CLASSROOMS_LIBRARY
vlan 99
 name MANAGEMENT
interface vlan 99
 ip address 192.168.22.75 255.255.255.248
 no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 4
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,99
! unused ports (final stage)
interface range fastEthernet0/5 - 23
 shutdown
interface range gigabitEthernet0/1 - 2
 shutdown
```

KATLEGO-SW2-LAB

```
hostname KATLEGO-SW2-LAB
vlan 20
 name DATA_COMPUTER_LAB
vlan 99
 name MANAGEMENT
interface vlan 99
 ip address 192.168.22.76 255.255.255.248
 no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 11
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 20,99
! unused ports (final stage)
interface range fastEthernet0/12 - 23
 shutdown
interface range gigabitEthernet0/1 - 2
 shutdown
```

KATLEGO-SW3-ADMIN

```
hostname KATLEGO-SW3-ADMIN
vlan 30
 name DATA_ADMIN
vlan 40
 name VOICE
vlan 99
 name MANAGEMENT
interface vlan 99
 ip address 192.168.22.77 255.255.255.248
 no shutdown
ip default-gateway 192.168.22.73
interface range fastEthernet0/1 - 5
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
interface fastEthernet0/6
 switchport mode access
 switchport access vlan 30
 switchport voice vlan 40
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
interface fastEthernet0/24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 30,40,99
! unused ports (final stage)
interface range fastEthernet0/7 - 23
 shutdown
interface range gigabitEthernet0/1 - 2
 shutdown
```

KATLEGO-RTR

```
hostname KATLEGO-RTR
interface gigabitEthernet0/0
 no shutdown
interface gigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.22.33 255.255.255.240
 ip access-group 100 in
interface gigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.22.1 255.255.255.224
 ip access-group 100 in
interface gigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.22.49 255.255.255.240
interface gigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.22.65 255.255.255.248
 ip access-group 100 in
interface gigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 192.168.22.73 255.255.255.248
interface gigabitEthernet0/1
 description Link to ISP
```

```
 ip address 203.0.113.2 255.255.255.252
 no shutdown
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip dhcp excluded-address 192.168.22.1
ip dhcp excluded-address 192.168.22.33
ip dhcp excluded-address 192.168.22.49 192.168.22.50
ip dhcp excluded-address 192.168.22.65
ip dhcp pool LAB_VLAN20
 network 192.168.22.0 255.255.255.224
 default-router 192.168.22.1
 dns-server 8.8.8.8
ip dhcp pool CLASSLIB_VLAN10
 network 192.168.22.32 255.255.255.240
 default-router 192.168.22.33
 dns-server 8.8.8.8
ip dhcp pool ADMIN_VLAN30
 network 192.168.22.48 255.255.255.240
 default-router 192.168.22.49
 dns-server 8.8.8.8
ip dhcp pool VOICE_VLAN40
 network 192.168.22.64 255.255.255.248
 default-router 192.168.22.65
 option 150 ip 192.168.22.65
access-list 100 remark Deny non-Admin VLANs from reaching the Server Room segment
access-list 100 deny ip any 192.168.22.72 0.0.0.7
access-list 100 permit ip any any
```

ISP-RTR and DNS-SERVER

```
hostname ISP-RTR
interface gigabitEthernet0/0
 description Link to Katlego
 ip address 203.0.113.1 255.255.255.252
 no shutdown
interface gigabitEthernet0/1
 description Link to DNS server
 ip address 8.8.8.1 255.255.255.0
 no shutdown
ip route 192.168.22.0 255.255.255.0 203.0.113.2
! DNS-SERVER (Server-PT): 8.8.8.8 255.255.255.0, gateway 8.8.8.1
! DNS service on; A record www.katlego.edu -> 8.8.8.8
```

The printer is set by hand: `192.168.22.50`, mask `255.255.255.240`, gateway `192.168.22.49`.

B Abbreviations

Term	Meaning	Term	Meaning
ACL	Access control list	NAT/PAT	(Port) address translation
ARP	Address Resolution Protocol	PoE	Power over Ethernet
DHCP	Dynamic Host Configuration Protocol	SVI	Switch virtual interface
DNS	Domain Name System	TTL	Time to live
FR	Functional requirement	VLAN	Virtual local area network
MAC	Media access control (hardware address)	VLSM	Variable Length Subnet Masking
CR	Change request	VoIP	Voice over IP